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State litigation is preserving federalism.

Shorman 26 [Jonathan Shorman, Covers democracy for States Newsroom, 04-03-2026, A New Era of Federal Power Tests States' Authority, Governing, https://www.governing.com/policy/a-new-era-of-federal-power-tests-states-authority //Lunde]

As the United States of America marks its 250th anniversary this year, the relationship between the states and the federal government is approaching a **breaking point**.

Led by a bellicose president, the executive branch has moved to dominate states, resulting in more than a year of escalating confrontations between the two levels of government.

President Donald Trump has worked quickly: In the first year of his second term, he surged thousands of **immigration** enforcement agents into a resistant Minneapolis and other cities, with fatal results. He seized control of the National Guard in some states against the will of governors.

His administration is trying to force states to turn over sensitive data on **millions of voters** ahead of the midterms. And it is blocking states from receiving, and distributing to their residents, billions of federal dollars for child care, public health, housing and a host of other congressionally **approved programs**.

Political parties have swung in and out of power in Washington for centuries, and recent administrations have increasingly clashed with states run by the other party. This time is different, dozens of sources in and around government told Stateline.

Trump and a coterie of loyal aides have set out to remake the nation in the president's image. Along the way, retribution and raw **power** have become the administration's primary tools to bend recalcitrant states to its will. Grants are pulled, armed force deployed, disaster aid withheld.

The states have repeatedly gone to court, asking the federal judiciary to rein in the executive branch. They have also started testing the bounds of their own authority, such as moving to restrict the actions of federal immigration enforcement agents.

The past year has led to a period of sustained state and federal conflict without parallel in modern U.S. history. The consequences for Americans over time will prove enormous, shaping the very nature of our government.

"This kind of battle between the federal government and the states, we've just never seen that before and it makes no sense," said former New Jersey Gov. Christine Todd Whitman, who was elected as a Republican but later helped co-found the centrist Forward Party.

Tensions between the states and the central government are as old as the nation itself. Alexander Hamilton famously favored a strong central government, while James Madison offered the Bill of Rights — including what became the 10th Amendment, which reserves for the states and the people those powers not delegated to the federal government.

But current strains are testing the bedrock principles of **federalism**, the uniquely American system created by the framers of the Constitution of power sharing between Washington, D.C., and the states.

Ahead of the 250th anniversary of the country's founding on July 4, Stateline is exploring how the Trump era is transforming the relationship between the states and the federal government. This article is the first in an occasional series, The 50 vs. The One, that will examine the current fraught moment and what evolving — and often deteriorating — state-federal ties mean for the country, now and in the future.

In interviews and public remarks, current and former elected officials at all levels of government, as well as experts on American government, have described the country as approaching a pivot point. Trump's second term could mark a defining moment for American federalism, one that will be studied in history books alongside Reconstruction, the New Deal and the Civil Rights Movement.

The United States will either continue to adhere to the principles of federalism, they say, or it will take a significant step toward a more powerful central government that sidelines the states.

“We are in a period of challenged federalism,” said Lisa Parshall, a federalism researcher and political science professor at Daemen University near Buffalo, New York. “The fact that we’re here talking about federalism tells you something about the current state of American politics.”

A federal agent lobs a tear gas canister toward protesters as agents advance through clouds of tear gas during clashes following the fatal shooting of a protester by immigration agents earlier in the day, on Jan. 24 in Minneapolis. It was the second fatal shooting of a civilian in the city, sparking fresh protests and outrage from state officials. The death of Alex Pretti came less than three weeks after another U.S. citizen, Renee Good, was shot and killed by an Immigration and Customs Enforcement officer involved in sweeps to round up undocumented migrants.

Dramatic Changes in a Year

Fears of diminishing state authority have animated state officials over the past year. Republican lawmakers in Utah have invested in federalism education and expanded a group to assess state-federal boundaries, for instance.

In July, Minnesota Gov. Tim Walz and Kansas Gov. Laura Kelly, both Democrats, publicly abandoned the nonpartisan National Governors Association, in part because they said the organization was not doing enough to protect states’ rights.

States are “**laboratories of democracy**,” Kelly said during an interview in February, using a classic civics textbook description. States have traditionally operated with relative freedom to pursue their own agendas and solutions to the challenges they face. In turn, states learn from one another.

“That’s been the beauty of it,” Kelly said. “If that’s to go away, if the federal government were — and they are, at this point — undermining states’ authority and responsibility, I think you end up slowing down the **entire country**.”

In the same way the three branches of government — the legislative, the executive and the judicial — provide checks and balances on one another, federalism imposes a state check on federal power. The U.S. Constitution, which went into effect in 1789, ensured states would command broad power over local commerce, policing, elections and other matters within their borders.

But Trump has at times raised doubt about whether he will always follow the Constitution and has claimed that he can ignore some of its requirements.

Last spring, Trump replied “I don’t know” when asked whether he needed to uphold the U.S. Constitution in the context of due process for immigrants. In 2022, he said massive election fraud allows parts of the Constitution to be terminated. And after his 2020 election defeat, he urged then-Vice President Mike Pence not to certify the results, even though the vice president has no constitutional authority to do so.

In February, Trump asserted that “states are just an agent of the federal government” as he called to “nationalize” elections. Under the Constitution, the responsibility of running elections belongs to the states.

Trump’s critics fault the Republican-controlled Congress for failing to challenge his sweeping assertions of executive power. His administration’s efforts to withhold from states billions in dollars appropriated by Congress, for instance, have spurred relatively little outrage among GOP lawmakers.

“What I think we’re seeing now is a whole different system of crushing state and local government,” said U.S. Rep. Emanuel Cleaver, a Missouri Democrat who has been in Congress since 2005. “And bowing down to a new system where we are almost living in a one-person government.”

In response to questions from Stateline, White House spokesperson Davis Ingle said in a statement: “The Trump Administration faithfully upholds our Constitution and the immortalized American principles of federalism, the rule of law, and the separation of powers.”

Trump and his allies have cast the president as a heroic figure capable of smashing through the machinery of government to achieve results on behalf of his voters and at the expense of his enemies. “For those who have been wronged and betrayed … I am your retribution,” he said in 2023.

He has at times taken steps that his supporters argue empower states, including effectively gutting the U.S. Department of Education, which Republicans have long accused of federal overreach. His appointments to the U.S. Supreme Court during his first term helped cement a

conservative majority that in 2022 returned the issue of abortion access to the states.

In a statement, the Republican Governors Association told Stateline the current administration trusts governors to run their own states.

“By cutting government bureaucracy and unnecessary red-tape, President Trump is empowering governors to make decisions that best serve their individual states,” wrote Kollin Crompton, an RGA spokesperson.

Scrambled Identities
The U.S. Constitution has been gradually amended in ways that have limited state power, most importantly through amendments that abolished slavery, required states to treat their citizens equally under the law, and prohibited states from denying suffrage on the basis of race and sex.

The federal government has also expanded its reach through legislation. President Franklin Roosevelt’s New Deal in the 1930s and President Lyndon Johnson’s Great Society in the 1960s imposed new economic regulations and created a federal social welfare apparatus that touches nearly every American.

Over time, Democrats broadly came to be seen as the party more comfortable with an active federal government and Republicans as the party seeking a more restrained Washington.

But the Trump era has scrambled those identities.

Trump has shown less respect for traditional conservative ideology, such as limited government and a general deference to the authority of states. Instead, he has taken a maximalist approach to executive power.

His actions have placed Democratic state officials in a position of **advancing limits** on the federal government, whether through **lawsuits** or legislation. And they have put Republican supporters of the president at odds with decades of conservative rhetoric.

“I do think that progressives are seeing that federalism — there’s a reason it’s in our constitutional order and it isn’t just something that’s left for conservatives,” said Sean Beienburg, an associate professor at Arizona State University who researches federalism and constitutional law.

In Los Angeles, Chicago and Portland, Oregon, Trump deployed federalized National Guard troops onto city streets before courts held him back and he withdrew. For a time, active-duty Marines also patrolled Los Angeles, an extraordinary use of the military for domestic purposes.

Oregon Democratic Attorney General Dan Rayfield, who challenged the deployment of the National Guard in his state, said the fight underscores why lawsuits matter in checking Trump’s power.

“People should be shocked that Oregon has filed 55 lawsuits,” Rayfield said in an interview earlier this year. “Their mind should be blown. But their mind should be equally blown at how often we’re winning these cases.”

The Trump administration has won **seven** court decisions — and lost **58** — so far, according to a New York Times litigation tracker.

Democratic state lawmakers have also searched for ways to restrict federal immigration agents. In California, Democratic Assemblymember Alex Lee has proposed prohibiting state tax breaks for Immigration and Customs Enforcement contractors — a move that could carry national implications because of the size of the state’s economy.

“We also, now, are **reasserting** what the role of the states and the federal government are,” Lee said.

States regulate data centers now but a national moratorium would erode federalism and spillover.

Huddleston 26 [Jennifer Huddleston, Senior fellow in technology policy at the Cato Institute, 07-01-2026, Turning a Local Issue into a Federal Headache: The Problem with Data Center Moratoriums, Cato Institute, https://www.cato.org/blog/turning-local-issue-federal-headache-problem-data-center-moratoriums //Lunde]

A new anti-data-center push seems to be uniting the left and the right, with critics ranging from Sen. Bernie Sanders and Rep. Frank Pallone to Sen. Josh Hawley and Rep. Nancy Mace calling for some form of federal limitation on data center construction. As with any expansion, there are legitimate policy conversations to have about resource use and other underlying issues. However, a moratorium on the construction of a particular kind of facility based on arbitrary ceilings when demand is growing is a misguided **policy sledgehammer** that could not only derail critical innovation but also fail to resolve the underlying policy issues.

Data centers aren't new, but the rising backlash and policy proposals to ban construction come at a critical time for innovation. The need for data centers should not be seen as an either/or but rather as an opportunity to consider better solutions to policy questions around resource use and other related issues facing the communities where they are built. If we aren't careful, bad policies like data center construction moratoriums and outdated energy policy could backfire — not only on progress in artificial intelligence, but also on a range of current and future technologies, including cloud computing and quantum computing.

The Rise of Data Center Construction Moratorium Policy and Its Problems

The most concrete federal effort to block data center construction is the AI Data Center Moratorium Act. Proposed by Senator Bernie Sanders on March 25, 2026, this Act would halt construction of new or existing data centers until a comprehensive federal AI regulatory framework is enacted. In the months since, the Overton window has shifted, with both the public and policymakers seemingly expressing a desire to pause construction of critical infrastructure for current and emerging technology demands.

The federal government is not alone. As of the end of June 2026, 116 municipalities have imposed **local moratoriums**. At the state level, California, Minnesota, Montana, and Kansas have all passed legislation that either incentivizes or disincentivizes data center construction in their states. These are in addition to the countless AI bills proposed in **state legislatures** across the country.

Unsurprisingly, states and localities have been and will continue to both **restrict** and enable datacenter construction for the foreseeable future.

Responding to Critics' Common Concerns

Senator Sanders justified his bill by arguing two main points common among data center critics. His first claim is that AI is advancing too quickly and that Congress is moving too slowly. Because of this, Congress needs a data center moratorium to give it time to catch up with the technology. His second claim is that data centers pose environmental and economic risks to their locality by raising utility rates and consumption.

The data on data centers, however, do not bear out these claims. Data centers are needed to meet increasing demands beyond just AI, including the volume of connected devices and data Americans use regularly. Our capacity needs have expanded with AI, but demand had been growing even before AI's exponential leaps. Data centers are critical to AI and the progress it enables across a variety of areas. AI products power not only the direct-to-consumer chatbots many have come to rely on to improve access to information but also an array of products that can improve health, enhance cybersecurity, or support responses to natural disasters. All of this is to say AI's benefits are far from merely ephemeral.

When it comes to concerns about energy usage or utility rates, there are better solutions than the moratorium sledgehammer, and the data indicate such claims are vastly overblown. Multiple studies have found that while energy rates have increased, it's far from clear that data centers are to blame. In fact, one study even found a connection between lower retail energy rates and data centers.

The reality is that energy policy keeps the grid and our energy supply outdated, which can increase costs. A better solution is to pursue energy policy reform, as our colleague Travis Fisher has written, to enable greater energy abundance and innovation.

Similarly, water usage in some regions of the country is a legitimate debate. But it is a problem that is far broader than just data centers. In aggregate, golf courses use significantly more water than data centers but rarely face the same kind of backlash.

Why The Backlash Could Backfire

The reality is that technological advancement is overwhelmingly a force for good. Some communities don't want a new data center, while others may embrace the millions in tax revenue and hundreds of high-paying jobs that come with it. Individual communities are best positioned to know the specific trade-offs they face with any large investment, as they have been for decades.

Data centers date back about 70 years to when IBM was building them to support the cutting-edge technology of the day: mainframe computers. The type of data centers we think of today can trace back over 25 years. Cities and regions like Loudon County, Virginia, have worked with communities and companies to find a balance that enables infrastructure

investments that provide economic opportunities and development while responding to community-specific concerns. More recently, however, these debates have spiraled from the typical debates over new construction or development to something much more heated.

The backlash against datacenters and AI has, at times, mushroomed into irrationality and violence. On April 10, 2026, a Molotov cocktail was thrown at Sam Altman's home. Two days later, the same home was struck by gunfire from a vehicle. In 2025, a 25-year-old was arrested and charged with an act of terrorism for his plan to burn down xAI's Memphis Colossus datacenter. These are paired with shootings at the homes of local councilmembers, physical confrontations at town meetings, and death threats against county commissioners.

This is particularly concerning, given the often-overlooked benefits of data centers. One study estimated that data center construction could create 4.7 million construction and skilled trade jobs in the United States. Many skilled trade jobs last beyond the initial construction, given the increased need for HVAC technicians and electricians, among other jobs, to support the center's operation.

For users, there is a growing demand for less latency and more storage. The downside is that without the necessary infrastructure to support the demands of modern startups, it grows more difficult to start the next great company in a garage. This could also mean that those needing this level of access to develop cutting-edge AI applications must migrate to regions that provide the necessary data center infrastructure to support it.

This is in part why a federal moratorium would be particularly concerning. Companies and talent need to know they will not face unnecessary roadblocks from the government, particularly for capital-intensive investments. Given the global nature of AI competition, preventing such a critical buildout could hinder the US's global competitiveness.

Conclusion

The debate over data centers has now jumped from a local issue to a national one. While the debate over AI policy should largely be a federal issue given its interstate nature, the issue of data centers is much more local and intrastate.

This should be concerning not only for its impact on infrastructure but also because a federal data center moratorium would represent an **extraordinary reach** by the federal government into **hyper-local** decisions about the development of individual plots of land across the country. Giving the **federal government** power over such a hyper-local issue might not be limited to data center construction and could **spill over** into an array of issues.

The goal should be to have a market that allows capital investment to increase supply and meet growing demands. Whether it's moratoriums that stop needed development or artificial inducements that could result in overbuilding or cronyism, state and federal government intrusion makes finding the actual correct balance more difficult for the local communities most directly impacted.

Banning data centers comes at a critical moment for AI. But they will impact much more, including not only the storage and speed we've come to expect in the cloud era, but also the advancements needed for the next generation of technology, such as quantum computing. The underlying concerns driving data center bans can be addressed with far better policy approaches that allow innovation while actually solving long-standing issues. Moratoriums on data center construction both fail to solve the concerns about resource usage and delay critical and potentially life-saving progress.

Strong federalism upholds democracy.

Galston 25 [William A. Galston, Senior Fellow – Governance Studies, Center for Effective Public Management (CEPM), Ezra K. Zilkha Chair in Governance Studies at Brookings, 9-5-2025, Professor Galston: US Federalism Slows the Shift Toward Competitive Authoritarianism, ECPS, https://www.populismstudies.org/professor-galston-us-federalism-slows-the-shift-toward-competitive-authoritarianism/ //Lunde]

In this incisive conversation, Brookings scholar Professor William A. Galston argues that America's decentralized system remains a **crucial brake** on executive overreach. While warning of real risks, he maintains, "We're not there yet," distinguishing the US from harder cases of institutional capture abroad. Professor Galston spotlights federalism and the courts as the decisive **arena of resistance**—urging institutions to defend their prerogatives through litigation, "not street protests but

the **law**." He assesses the influence of Project 2025, redistricting fights in Texas/California, and the politics of immigration, crime, and DEI, noting potential backlash among centrist voters. The result is a clear-eyed appraisal of democratic resilience—and the legal contests that will shape whether the US moves toward or away from competitive **authoritarianism**.

Interview by Selcuk Gultasli

Giving an in-depth interview to the European Center for Populism Studies (ECPS), Professor William A. Galston — senior fellow and the Ezra K. Zilkha Chair in the Governance Studies program at Brookings Institution and a leading authority on American governance, populism, and institutional resilience — provides a comprehensive analysis of the evolving dynamics of US democracy under President Donald Trump's second administration. Drawing on his deep expertise in political institutions and constitutional law, Professor Galston examines how federalism, legal contestation, and civil society remain central to safeguarding **checks and balances** amid mounting executive centralization.

As the title emphasizes, "US. Federalism Slows the Shift Toward Competitive Authoritarianism," Professor Galston underscores that America's decentralized political structure continues to serve as a buffer against executive overreach. He stresses that, despite concerns about democratic **backsliding**, the US is not yet on par with countries such as Hungary, Turkey, or India, where institutional capture has been far more extensive. "We're **not there yet**," Professor Galston explains, "and I am hopeful that we'll never get there."

A recurring theme throughout the interview is the growing role of law as the principal arena of political struggle. Professor Galston argues that the resilience of US democracy increasingly depends on institutions defending their **legal prerogatives**: "Pushing back, using not street protests but the law, will be the most important venue of contestation."

This dynamic is vividly illustrated in his discussion of a recent legal battle involving Harvard University, which challenged the Trump administration's attempt to withhold federal research funds. A federal judge sided with the university, ruling that the administration's actions were unconstitutional. For Professor Galston, this episode demonstrates that institutions "with the will to defend their legal rights can push back — and push back effectively."

The interview also delves into the strategic influence of Project 2025 and the Heritage Foundation in shaping Trump's second-term agenda. Professor Galston notes that the initiative has successfully translated its ideas into both institutional frameworks and personnel appointments, allowing the administration to pursue expansive interpretations of executive authority while testing the limits of constitutional constraints.

Additionally, Professor Galston examines redistricting battles, such as those in Texas and California, highlighting how federalism enables both parties to counterbalance each other's maneuvers, thereby slowing the concentration of power at the national level.

Finally, Galston reflects on the broader trajectory of US democracy, warning that the coming years will be decisive in determining whether **constitutional norms** can withstand mounting pressures. Yet he remains cautiously optimistic, noting that state governments, universities, professional associations, and civil society are "waking up" to the importance of defending democratic principles.

This interview offers a nuanced, scholarly perspective on the complex interplay between institutional resilience and executive ambition, providing crucial insights into America's democratic future.

Only a strong US democracy can maintain the present global order.

Kagan 12 [Robert Kagan, Stephen & Barbara Friedman Senior Fellow – Foreign Policy, Strobe Talbott Center for Security, Strategy, and Technology at Brookings, 2-11-2012, Why the World Needs America, Brookings, https://www.brookings.edu/articles/why-the-world-needs-america/ //Lunde]

History shows that world orders, including our own, are transient. They rise and fall, and the institutions they erect, the beliefs and "norms" that guide them, the economic systems they support—they rise and fall, too. The downfall of the Roman Empire brought an end not just to Roman rule but to Roman government and law and to an entire economic system stretching from Northern Europe to North Africa. Culture, the arts, even progress in science and technology, were set back for centuries.

Modern history has followed a similar pattern. After the Napoleonic Wars of the early 19th century, British control of the seas and the balance of great powers on the European continent provided relative security and stability. Prosperity grew, personal freedoms expanded, and the world was knit more closely together by revolutions in commerce and communication.

With the outbreak of World War I, the age of settled peace and advancing liberalism—of European civilization approaching its pinnacle—collapsed into an age of hyper-nationalism, despotism and economic calamity. The once-promising spread of democracy and liberalism halted and then reversed course, leaving a handful of outnumbered and besieged democracies living nervously in the shadow of fascist and totalitarian neighbors. The collapse of the British and European orders in the 20th century did not produce a new dark age—though if Nazi Germany and imperial Japan had prevailed, it might have—but the horrific conflict that it produced was, in its own way, just as devastating.

Would the end of the present American-dominated order have less dire consequences? A surprising number of American intellectuals, politicians and policy makers greet the prospect with equanimity. There is a general sense that the end of the era of American pre-eminence, if and when it comes, need not mean the end of the present international order, with its widespread freedom, unprecedented global prosperity (even amid the current economic crisis) and absence of war among the great powers.

American power may diminish, the political scientist G. John Ikenberry argues, but "the underlying foundations of the liberal international order will survive and thrive." The commentator Fareed Zakaria believes that even as the balance shifts against the U.S., rising powers like China "will continue to live within the framework of the current international system." And there are elements across the political spectrum—Republicans who call for retrenchment, Democrats who put their faith in international law and institutions—who don't imagine that a "post-American world" would look very different from the American world.

If all of this sounds too good to be true, it is. The present world order was largely shaped by American power and reflects **American interests** and preferences. If the balance of power shifts in the direction of other nations, the world order will change to suit their interests and preferences. Nor can we assume that all the great powers in a post-American world would agree on the benefits of preserving the present order, or have the capacity to preserve it, even if they wanted to.

Take the issue of **democracy**. For several decades, the balance of power in the world has favored **democratic governments**. In a genuinely post-American world, the balance would shift toward the great-power **autocracies**. Both Beijing and Moscow already protect dictators like Syria's Bashar al-Assad. If they gain greater relative influence in the future, we will see fewer **democratic transitions** and more autocrats hanging on to power. The balance in a new, multipolar world might be more favorable to democracy if some of the rising democracies—Brazil, India, Turkey, South Africa—picked up the slack from a declining U.S. Yet not all of them have the desire or the **capacity** to do it.

What about the economic order of free markets and free trade? People assume that China and other rising powers that have benefited so much from the present system would have a stake in preserving it. They wouldn't kill the goose that lays the golden eggs.

Unfortunately, they might not be able to help themselves. The creation and survival of a liberal economic order has depended, historically, on great powers that are both willing and able to support open trade and free markets, often with naval power. If a declining America is unable to maintain its long-standing hegemony on the high seas, would other nations take on the burdens and the expense of sustaining navies to fill in the gaps?

Even if they did, would this produce an open global commons—or rising tension? China and India are building bigger navies, but the result so far has been greater competition, not greater security. As Mohan Malik has noted in this newspaper, their "maritime rivalry could spill into the open in a decade or two," when India deploys an aircraft carrier in the Pacific Ocean and China deploys one in the Indian Ocean. The move from American-dominated oceans to collective policing by several great powers could be a recipe for competition and conflict rather than for a liberal economic order.

And do the Chinese really value an open economic system? The Chinese economy soon may become the largest in the world, but it will be far from the richest. Its size is a product of the country's enormous population, but in per capita terms, China remains relatively poor. The U.S., Germany and Japan have a per capita GDP of over $40,000. China's is a little over $4,000, putting it at the same level as Angola, Algeria and Belize. Even if optimistic forecasts are correct, China's per capita GDP by 2030 would still only be half that of the U.S., putting it roughly where Slovenia and Greece are today.

As Arvind Subramanian and other economists have pointed out, this will make for a historically unique situation. In the past, the largest and most dominant economies in the world have also been the richest. Nations whose peoples are such obvious winners in a relatively unfettered economic system have less temptation to pursue protectionist measures and have more of an incentive to keep the system open.

China's leaders, presiding over a poorer and still developing country, may prove less willing to open their economy. They have already begun closing some sectors to foreign competition and are likely to close others in the future. Even optimists like Mr. Subramanian believe that the liberal economic order will require "some insurance" against a scenario in which "China exercises its dominance by either reversing its previous policies or failing to open areas of the economy that are now highly protected." American economic dominance has been welcomed by much of the world because, like the mobster Hyman Roth in "The Godfather," the U.S. has always made money for its partners. Chinese economic dominance may get a different reception.

Another problem is that China's form of capitalism is heavily dominated by the state, with the ultimate goal of preserving the rule of the Communist Party. Unlike the eras of British and American pre-eminence, when the leading economic powers were dominated largely by private individuals or companies, China's system is more like the mercantilist arrangements of previous centuries. The government amasses wealth in order to secure its continued rule and to pay for armies and navies to compete with other great powers.

Although the Chinese have been beneficiaries of an open international economic order, they could end up undermining it simply because, as an autocratic society, their priority is to preserve the state's control of wealth and the power that it brings. They might kill the goose that lays the golden eggs because they can't figure out how to keep both it and themselves alive.

Finally, what about the long peace that has held among the great powers for the better part of six decades? Would it survive in a post-American world?

Most commentators who welcome this scenario imagine that American predominance would be replaced by some kind of multipolar harmony. But multipolar systems have historically been neither particularly stable nor particularly peaceful. Rough parity among powerful nations is a source of uncertainty that leads to miscalculation. Conflicts erupt as a result of fluctuations in the delicate power equation.

War among the great powers was a common, if not constant, occurrence in the long periods of multipolarity from the 16th to the 18th centuries, culminating in the series of enormously destructive Europe-wide wars that followed the French Revolution and ended with Napoleon's defeat in 1815.

The 19th century was notable for two stretches of great-power peace of roughly four decades each, punctuated by major conflicts. The Crimean War (1853-1856) was a mini-world war involving well over a million Russian, French, British and Turkish troops, as well as forces from nine other nations; it produced almost a half-million dead combatants and many more wounded. In the Franco-Prussian War (1870-1871), the two nations together fielded close to two million troops, of whom nearly a half-million were killed or wounded.

The peace that followed these conflicts was characterized by increasing tension and competition, numerous war scares and massive increases in armaments on both land and sea. Its climax was World War I, the most destructive and deadly conflict that mankind had known up to that point. As the political scientist Robert W. Tucker has observed, "Such stability and moderation as the balance brought rested ultimately on the threat or use of force. War remained the essential means for maintaining the balance of power."

There is little reason to believe that a return to multipolarity in the 21st century would bring greater peace and stability than it has in the past. The era of American predominance has shown that there is no better recipe for great-power peace than certainty about who holds the upper hand.

President Bill Clinton left office believing that the key task for America was to "create the world we would like to live in when we are no longer the world's only superpower," to prepare for "a time when we would have to share the stage." It is an eminently sensible-sounding proposal. But can it be done? For particularly in matters of security, the rules and institutions of international order rarely survive the decline of the nations that erected them. They are like scaffolding around a building: They don't hold the building up; the building holds them up.

Many foreign-policy experts see the present international order as the inevitable result of human progress, a combination of advancing science and technology, an increasingly global economy, strengthening international institutions, evolving "norms" of international behavior and the gradual but inevitable triumph of liberal democracy over other forms of government—forces of change that transcend the actions of men and nations.

Americans certainly like to believe that our preferred order survives because it is right and just—not only for us but for everyone. We assume that the triumph of democracy is the triumph of a better idea, and the victory of market capitalism is the victory of a better system, and that both are irreversible. That is why Francis Fukuyama's thesis about "the end of history" was so attractive at the end of the Cold War and retains its appeal even now, after it has been discredited by events. The idea of inevitable evolution means that there is no requirement to impose a decent order. It will merely happen.

But international order is not an evolution; it is an **imposition**. It is the domination of one vision over others—in America's case, the domination of free-market and **democratic principles**, together with an international system that supports them. The present order will last only as long as those who favor it and benefit from it retain the **will** and capacity to **defend it**.

There was nothing inevitable about the world that was created after World War II. No divine providence or unfolding Hegelian dialectic required the triumph of democracy and capitalism, and there is no guarantee that their success will outlast the powerful nations that have fought for them. Democratic progress and liberal economics have been and can be reversed and undone. The ancient democracies of Greece and the republics of Rome and Venice all fell to more powerful forces or through their own failings. The evolving liberal economic order of Europe collapsed in the 1920s and 1930s. The better idea doesn't have to win just because it is a better idea. It requires great powers to champion it.

If and when American power declines, the institutions and norms that American power has supported will decline, too. Or more likely, if history is a guide, they may collapse altogether as we make a transition to another kind of world order, or to disorder. We may discover then that the U.S. was essential to keeping the present world order together and that the alternative to American power was not peace and harmony but chaos and catastrophe—which is what the world looked like right before the American order came into being.

Strong democracy caps conflict escalation and existential crises.

Merkel 26 [Katharina Merkel, Katharina Merkel is an independent research and evaluation consultant specialising in governance, security, and development policy. She has over 16 years of experience leading research and evaluations across multiple regions, working with international organisations, governments, and civil society actors, May 2026, "Democracy as a Security Asset", WFD, https://www.wfd.org/sites/default/files/2026-06/JN_0189%20wfd-democracy-security-asset-global-evidence-review-2026-05%20AW.pdf //Lunde]

3. Democratic Constraints on Interstate Escalation

Headline finding:

Democratic institutional arrangements strengthen interstate security by embedding decisions on the use of force within systems of scrutiny, accountability, and shared authority that raise the political costs of unnecessary escalation.

Strength of evidence: Moderate–Strong

A substantial body of comparative research examines how democratic institutional arrangements influence decisions to initiate and escalate interstate conflict. Building on the broader democratic peace literature, which finds that democracies are significantly less likely to engage in violent **conflict** with one another, this research moves beyond regime type comparisons to investigate how domestic political structures shape executive decision-making under conditions of crisis and uncertainty. Across cross-national and dyadic analyses, democratic systems are associated with higher **procedural** and political **thresholds** for the use of force (Reiter and Tillman 2002; Palmer, London and Regan 2004). Legislative oversight and active **political opposition** embed security decisions within systems of scrutiny and contestation, making rapid or unilateral escalation **less likely** (Howell and Pevehouse 2007; Dieterich, Hummel and Marschall 2015). Democratic institutions do not eliminate conflict behaviour, but they alter how leaders calculate risks by requiring broader political support for military action and by exposing security decisions to public debate.

One of the central mechanisms identified in this literature is the role of domestic political competition in revealing information during international crises. Research on coercive diplomacy shows that opposition actors in democratic systems can publicly challenge executive claims about threats or military necessity, thereby signalling domestic constraints to foreign adversaries and reducing incentives for bluffing or miscalculation (Schultz 1998; see also Prins 2003; Prins and Sprecher 1999). Political competition therefore functions as an information mechanism that encourages more cautious escalation strategies. When leaders must justify security policies to both **domestic audiences** and political rivals, they face greater scrutiny regarding the credibility of threats and the potential costs of military action, which can slow escalation and create additional opportunities for **diplomatic** bargaining.

“Democratic institutions raise the political costs of war by subjecting decisions on the use of force to scrutiny and accountability”.

Evidence from studies of military intervention behaviour further supports the importance of institutional constraints in shaping external security policy. Cross-national analyses indicate that democratic leaders tend to exercise greater selectivity when initiating military conflicts and face higher political thresholds for the use of force (Reiter and Tillman 2002; Palmer, London and Regan 2004). Electoral accountability plays a central role in this dynamic. Leaders operating within democratic systems must anticipate the domestic political consequences of military failure, which increases the risks associated with poorly justified or unsuccessful military interventions. Legislative involvement in authorising the use of force can further increase the political costs of escalation by requiring executives to secure approval from additional veto players before initiating military action (Howell and Pevehouse 2007; Dieterich, Hummel and Marschall 2015).

A related strand of research highlights how political institutions shape the incentives facing leaders themselves. Comparative studies show that leaders’ decisions regarding war and peace are strongly influenced by the political environments in which they operate, with authoritarian leaders facing different constraints and incentives for risk-taking than their democratic counterparts (Weeks 2012). Research on leader characteristics similarly finds that individual traits such as prior experience can influence conflict behaviour, but that these effects are conditioned by broader institutional contexts that shape how leadership decisions translate into policy outcomes (Horowitz and Stam 2014).

Taken together, this literature suggests that democratic institutions influence interstate security behaviour not by eliminating geopolitical competition but by structuring the decision-making environments in which leaders operate. Political competition, legislative oversight, and public scrutiny raise the political costs associated with risky military behaviour and embed security decisions within broader processes of deliberation and accountability. Leadership behaviour therefore constitutes one of the channels through which democratic institutional constraints translate into observable security outcomes, shaping patterns of crisis escalation, conflict initiation, and the strategic use of military force.

4. Information Integrity and Political Polarisation

Headline finding:

Democratic institutions strengthen information resilience by enabling contestation, transparency, and institutional trust, limiting the security impact of information manipulation despite greater exposure to influence operations.

Strength of evidence: Moderate

A growing body of comparative research examines how democratic systems perform under conditions of information manipulation and foreign influence operations. Democracies are structurally more exposed to such activities because openness, civil liberties, and competitive elections create accessible entry points for external actors seeking to influence public debate. Open media systems allow rapid dissemination of information, while plural political competition increases the strategic value of shaping narratives and political agendas. Conceptual and empirical work therefore consistently identifies democratic societies as attractive targets for influence operations and coordinated disinformation campaigns (Bradshaw and Howard 2019; Whyte 2020). Exposure, however, does not automatically translate into systemic destabilisation.

Empirical research across multiple countries and platforms finds limited evidence that foreign disinformation produces large-scale persuasion or durable shifts in political attitudes.

“Plural media and political competition make information systems more resilient to manipulation”.

Linked survey and social media data analyses show that exposure to foreign propaganda typically affects relatively small subsets of politically engaged users and remains marginal compared with the influence of domestic political communication (Bail et al. 2020; Eady et al. 2023). Large-scale systematic reviews similarly conclude that robust causal evidence for widespread persuasion effects remains limited, with disinformation campaigns more reliably affecting political attention, agenda-setting, and polarisation dynamics than deeply held beliefs (Lorenz-Spreen et al. 2023). These findings suggest that the primary risks posed by information manipulation lie less in wholesale opinion change than in the amplification of existing political divisions, increased distrust in institutions, and shifts in public attention that can intensify polarisation within democratic societies. Comparative research also indicates that vulnerability to misinformation depends heavily on social and cognitive factors rather than regime type alone. Cross-national measurement studies show substantial variation in susceptibility to misinformation across individuals and countries that cannot be explained simply by differences in democratic institutions (Maertens et al. 2024; Kyrychenko et al. 2025). Institutional trust emerges as a key mediating variable.

Comparative surveys find that lower levels of trust in democratic institutions are associated with greater willingness to share misleading information, while higher levels of institutional confidence correlate with more accurate information evaluation and reduced misinformation sharing (Humprecht et al. 2023; Hameleers, Brosius and de Vreese 2022). Behavioural evidence further suggests that belief in false information tends to be concentrated among politically engaged minorities rather than diffused across entire populations (Guess et al. 2020).

The relationship between democracy and information security is therefore characterised by a dual dynamic. Democratic openness increases exposure to manipulation attempts, but democratic institutions also generate important resilience mechanisms. Open media, plural political competition, and public scrutiny create environments in which misleading information can be contested and corrected. The stability of democratic information environments depends heavily on the credibility of institutions and the level of trust citizens place in them. When

institutional legitimacy remains high, open public spheres can function as self-correcting systems in which competing sources of information limit the long-term influence of manipulation campaigns. In this sense, democracy does not eliminate informational threats but shapes how societies respond to them, balancing structural exposure with institutional resilience.

5. Cyber Conflict Behaviour and Cybersecurity Governance

Headline finding:

Democratic institutions shape patterns of cyber conflict by encouraging restraint and accountability in offensive operations, while cybersecurity capacity depends primarily on governance effectiveness and institutional capability rather than regime type alone.

Strength of evidence: Emerging–Moderate

Emerging research on cyber conflict extends the democracy–security relationship into domains characterised by persistent strategic competition below the threshold of conventional warfare. Cyber operations allow states to conduct espionage, signalling, and disruption without triggering open armed confrontation, making them a particularly revealing arena for examining how political institutions shape security behaviour. Comparative studies do not suggest that democracies abstain from cyber competition. Instead, the evidence indicates that democratic systems display distinct patterns in the initiation, targeting, and escalation of cyber operations that reflect institutional accountability, legal constraints, and reputational considerations. Cross-national quantitative analyses show that states with higher levels of political rights are less likely to initiate publicly observable cyberattacks, even after controlling for cyber capability and geopolitical context (Hunter, Albert and Garrett 2021). Panel analyses similarly find that democratic governance remains associated with lower probabilities of cyber initiation once differences in power, technological capacity, and external threat environments are taken into account (Hunter et al. 2022).

Institutional constraints appear to shape cyber behaviour primarily by raising the political and legal costs of authorising offensive operations. Democratic governments typically operate within systems of bureaucratic oversight, legal review, and legislative scrutiny that increase the risk of political backlash if cyber operations produce unintended escalation or civilian harm. As a result, democratic states tend to employ cyber capabilities selectively and often favour limited operations designed for signalling, intelligence gathering, or disruption rather than large-scale destructive effects. Comparative analyses of global cyber incident datasets suggest that cyber operations are frequently structured to remain bounded and reversible, limiting escalation risks and maintaining plausible deniability (Valeriano, Jensen and Maness 2018). Research examining cyber activity during armed conflict likewise finds that cyber tools are often used to complement rather than escalate conventional operations, serving signalling and intelligence functions rather than triggering uncontrolled military confrontation (Lindsay 2013; Kostyuk and Zhukov 2019). Evidence at the dyadic level suggests that democratic restraint in cyberspace operates conditionally rather than universally. Democracies do conduct cyber operations, yet they rarely target other democracies, mirroring patterns observed in conventional democratic peace research and peace theories. Dyadic quantitative analyses identify significantly lower levels of cyber hostility between democratic pairs, while cyber competition remains directed primarily toward authoritarian or mixed-regime adversaries (Hunter et al. 2024; Geiger 2021). Democratic cyber behaviour also differs in the domain of attribution and public signalling. Decisions to publicly attribute cyber incidents are often shaped by domestic audience costs and reputational concerns, leading democratic governments to frame cyber responses through legal narratives and coalition-building rather than relying exclusively on covert retaliation (Poznansky and Perkoski 2018).

“Cyber operations in democratic states are more likely to be politically constrained and strategically calibrated”.

A related but analytically distinct literature examines whether democratic governance influences cybersecurity capacity itself. In contrast to cyber conflict behaviour, comparative evidence suggests that regime type alone is a weak predictor of national cyber resilience. Cross-national analyses consistently show that cybersecurity outcomes correlate more strongly with governance effectiveness, administrative coordination, regulatory coherence, and sustained investment in technical expertise than with democratic participation alone (Creese et al. 2021; Chen et al. 2023). Institutional stability and state capacity appear to be decisive factors shaping national cybersecurity preparedness, with both consolidated democracies and stable authoritarian systems often outperforming hybrid or weakly institutionalised regimes (Kleiner 2025). Studies of national cybersecurity strategy adoption likewise highlight the importance of international cooperation and institutional diffusion mechanisms, including membership in international organisations and participation in global governance networks (Kostyuk and Sidorova 2026). Cybersecurity therefore illustrates an important qualification within the democracy–security nexus: democratic institutions influence how cyber competition is conducted, but effective cyber resilience depends primarily on governance capability and institutional performance rather than regime type alone.

6. Crisis Response, Institutional Adaptation, and Long-Term Resilience

Headline finding:

Democratic institutions strengthen security by reducing the risk that crises **escalate** into large-scale instability, through greater transparency, politically incentivised response, and societal compliance.

Strength of evidence: Moderate

A substantial comparative literature examines how political institutions shape the ability of states to manage both acute shocks and slower-moving structural pressures that can generate security risks, including natural disasters, **pandemics**, **environmental degradation**, and **resource scarcity**. Across these domains, the most consistent finding is not that democracies prevent crises, but that they alter the pathways through which crises unfold. Democratic governance reduces the probability that emerging risks escalate into severe humanitarian or political breakdown by embedding political competition, public scrutiny, and **institutionalised accountability** into decision-making processes. These features shape how information about risks is generated and contested, how governments respond to emerging pressures, and how the political costs of inaction are distributed. In particular, democracies tend to operate in more transparent information environments, disclosing more economic and governance-relevant data and reducing incentives to conceal adverse developments (Hollyer, Rosendorff and Vreeland 2011; Williams 2015). Independent media and political opposition further increase the visibility of emerging risks, making them more likely to become publicly salient at **earlier stages**, when intervention may still be **feasible**. While these dynamics do not guarantee early detection, they strengthen the informational basis for policy responses and increase the likelihood that emerging threats are subject to scrutiny and debate rather than suppressed or ignored.

These informational conditions translate into distinct political incentives for crisis response, particularly where failure to act produces visible human costs. Comparative evidence from disaster governance shows that democracies experience significantly lower mortality from natural disasters, even after controlling for income levels and geographic exposure (Kahn 2005), while research on famine demonstrates that no substantial famine has occurred in a functioning democracy, a pattern widely attributed to the interaction of electoral accountability and press freedom (Sen 1981; 1999). In these contexts, the causal mechanism lies in the political consequences of inaction: leaders subject to electoral competition face direct risks when crises become visible, while opposition actors and media institutions amplify failures of response. These pressures encourage earlier intervention, investment in mitigation and preparedness, and more effective mobilisation during crises (Keefer, Neumayer and Plümper 2011; Raschky 2008). Similar incentive structures operate in environmental governance. Democracies are more likely to participate in international environmental agreements and adopt climate-related policies, reflecting the influence of electoral incentives, public scrutiny, and organised civil society actors (Neumayer 2002; Bättig and Bernauer 2009). Political competition can increase the costs of environmental degradation, encouraging governments to adopt regulatory frameworks that address long-term risks even where short-term economic incentives favour inaction (Fredriksson 2005). At the same time, the relationship between democracy and environmental outcomes remains conditional. While some studies identify associations between democracy and improvements in environmental quality, these effects vary according to pollutants, levels of development, and enforcement capacity (Li and Reuveny 2006; Bernauer and Koubi 2009). This variation reflects the complexity of environmental governance, which involves long time horizons, diffuse costs, and collective-action problems. Democratic institutions do not consistently produce superior environmental outcomes, but they shape the processes through which environmental risks are managed by facilitating regulatory adoption, sustaining public debate over trade-offs, and enabling policy adjustment over time.

A further mechanism operates through societal compliance and cooperation, which are critical for effective crisis management. Empirical evidence shows that higher levels of political trust are associated with greater voluntary compliance with government directives. Mobility data from Europe during the COVID-19 pandemic indicate that higher political trust is associated with stronger adherence to mobility restrictions (Bargain and Aminjonov 2020), while observational studies show that higher social trust is associated with stronger compliance with mitigation measures (Goldstein and Wiedemann 2021). These dynamics are directly relevant to both acute and long-term risks, including environmental governance, where mitigation and adaptation depend heavily on sustained public cooperation. Democratic legitimacy, by increasing trust in institutions and the perceived fairness of policy decisions, enhances governments' ability to mobilise such cooperation without relying exclusively on coercion. Evidence from the COVID-19 pandemic further illustrates how democratic institutions shape crisis dynamics through transparency and accountability. Cross-national analyses show that democracies exhibited higher reported mortality in the early stages of the pandemic, partly due to more transparent reporting and differences in exposure and policy timing (Cepaluni, Dorsch and Branyiczki 2020), while differences in reported mortality between regime types are partly driven by underreporting in less democratic systems (Annaka 2021). Early cross-national analyses based on first-wave data found no consistent regime-type advantage in pandemic containment effectiveness (Cheibub, Hong and Przeworski 2020), while later cross-national studies using excess mortality similarly find that differences between regime types are limited and strongly mediated by reporting practices and state capacity (Annaka 2021). Some evidence further indicates that democratic accountability may be associated with lower mortality when accounting for reporting differences and broader measures of mortality, reflecting the role of public scrutiny in encouraging policy adjustment over time (Frey, Chen and Presidente 2020).

Overall, the comparative evidence indicates that democratic governance contributes to security primarily by reshaping how crises, both sudden and slow-moving, are managed over time. Democratic institutions increase the likelihood that emerging risks become visible, that governments face incentives to respond, and that societies comply with measures required to mitigate harm. They also enable policy correction and institutional adjustment, reducing the probability that initial failures escalate into systemic instability. The security advantages of democracy therefore lie less in preventing shocks altogether than in strengthening the capacity of political systems to manage, absorb, and adapt to crises without triggering large-scale humanitarian or political breakdown.

Our metastudy analyzes two centuries of data.

Merkel 26 [Katharina Merkel, Katharina Merkel is an independent research and evaluation consultant specialising in governance, security, and development policy. She has over 16 years of experience leading research and evaluations across multiple regions, working with international organisations, governments, and civil society actors, May 2026, "Democracy as a Security Asset", WFD, https://www.wfd.org/sites/default/files/2026-06/JN_0189%20wfd-democracy-security-asset-global-evidence-review-2026-05%20AW.pdf //Lunde]

METHODOLOGY

The approach taken in this review is explicitly empirical, comparative, and mechanism-focused. It draws on a structured **synthesis** of high-quality, cross-national quantitative studies, quasi-experimental analyses, panel datasets, and experimental research where available. The underlying evidence base spans a wide temporal range, from long-run historical panel data beginning in the early **19th century** to contemporary studies using data from the 2010s and 2020s, with the majority of analyses drawing on post-1945 and especially post-1960 cross-national datasets. The objective is not to assemble illustrative examples, but to identify recurring empirical patterns. Across domains, the review examines whether findings converge when tested using alternative datasets, time periods, and measurement strategies.

The review prioritises **peer-reviewed** academic research and does not systematically include grey literature. This reflects a focus on studies that provide transparent methods, replicable data, and clearly specified empirical strategies, which are necessary for assessing robustness across datasets, timeframes, and measurement approaches. While policy reports and other grey literature can offer valuable contextual insights, they often rely on case-based evidence or lack sufficient methodological detail to support systematic comparative analysis and consistent evaluation of causal claims.

Evidence is assessed separately for (1) the impact of democratic processes and institutions on security outcomes and (2) the impact of security pressures on democratic governance. To capture these relationships more precisely, the review disaggregates both democracy and security into their component dimensions rather than relying solely on aggregate regime or security indices.

Democracy is examined through specific institutional and societal features, including the rule of law, judicial independence (both de jure and de facto), legislative oversight, media freedom, accountability mechanisms, civil–military relations, political participation, inclusion, and gender equality.

Security is treated as a multi-dimensional concept encompassing internal conflict and political violence, state repression, terrorism, interstate hostility, cyber operations, hybrid interference, crisis governance, and broader dimensions of human and societal security such as social cohesion, gender-based violence, and environmental security.

This disaggregation allows for a more precise identification of the institutional and societal mechanisms linking democracy and security and helps avoid overgeneralisation from broad regime categories.

Causal inference:

The review distinguishes between different levels of causal inference across the studies included in the evidence base. Experimental studies are treated as providing the highest level of internal validity, although such designs are rare in research on macro-institutional and cross-national political phenomena. Where relevant, experimental studies conducted within individual country contexts are included because they can provide insights into behavioural mechanisms. Quasi-experimental designs, such as natural experiments, instrumental variable approaches, and

difference-in-differences analyses, are treated as providing the second-highest level of causal inference. Panel analyses with fixed effects are used to identify recurring cross-national patterns while accounting for country-specific characteristics over time, though they cannot fully eliminate concerns about omitted variables or simultaneity. Purely observational cross-national correlations are included where theoretically relevant but are not interpreted as causal without supporting theoretical justification. Where the direction of causality remains uncertain, this uncertainty is explicitly acknowledged.

Assessment of Evidence Strength:

Studies included in the review are first assessed individually according to the evidentiary framework outlined above, with detailed results presented in the evidence tables in Annex A, and then considered collectively when evaluating the strength of evidence for specific relationships examined in the review. Rather than relying on single studies, the analysis focuses on patterns across the **body of literature**, examining whether findings converge across different datasets, research designs, time periods, and measurement strategies. The strength of evidence for each domain is therefore determined by the cumulative weight of available research, including the number of studies addressing the relationship, the methodological rigour of those studies, and the degree of consistency in their findings.

Lobbying decks it.

Rodskog 26 [Thomas Rodskog (Founder of AI Safety @ UCSB, Student @ UCSB), 03-15-2026, "Sanders's Data Center Moratorium Is Risky Strategy for AI Safety", LessWrong, https://web.archive.org/web/20260520222255/https://www.lesswrong.com/posts/GSD8bEjREYioBDisr/sanders-s-data-center-moratorium-is-risky-strategy-for-ai, DOA: 7-9-2026]|eti

Even in the best case for this legislation, a data center moratorium is a temporary measure. Without concrete political pressure for a pause on superintelligence, there are just too many forces fighting against this type of legislation. National security is, rightfully, an important concern in our political climate. AI development must continue to keep up with China's military capabilities. The entire economy is also betting on AI-powered growth. There are extremely **well-funded actors**, from the major AI labs to the venture capital firms backing them, that will lobby aggressively against any construction freeze, and they have far more **political leverage** than Sanders does. Without broad public support, which is building but not here yet, this thing will get **repealed** (assuming it could even pass).

And even while a moratorium is in effect, it only targets one part of the capabilities pipeline. Existing data centers **keep running**, and capital will be **reallocated** to do better research with existing compute, or build up energy infrastructure in preparation for when data center construction can resume. The labs don't just sit around. This means that even a full year of a construction freeze (which seems wildly unlikely given the current administration) might only delay frontier AI capabilities by a matter of months.

Extinction!

Clare 23 [Stephen Clare, MsC in natural resource management and policy + Applied Researcher @ Founders Pledge + a Program Analyst for UNDP + writer and existential risks researcher @ 80,000 Hours, 6-xx-2023, Great power war, 80,000 Hours, https://80000hours.org/problem-profiles/great-power-conflict/]

A modern great power war could see **nuclear weapons**, **bioweapons**, **autonomous weapons**, and other destructive **new technologies** deployed on an **unprecedented** scale.

It would probably be the most destructive event in history, shattering our world. It could even threaten us with extinction.

We've come perilously close to just this kind of catastrophe before.

On October 27, 1962 — near the peak of the Cuban Missile Crisis — an American U-2 reconnaissance plane set out on a routine mission to the Arctic to collect data on Soviet nuclear tests. But, while flying near the North Pole, with the stars obscured by the northern lights, the pilot made a navigation error and strayed into Soviet airspace.1

Soviet commanders sent fighter jets to intercept the American plane. The jets were picked up by American radar operators and nuclear-armed F-102 fighters took off to protect the U-2.

Fortunately, the reconnaissance pilot realised his error with enough time to correct course before the Soviet and American fighters met. But the intrusion enraged Soviet Premier Nikita Khrushchev, who was already on high alert amidst the crisis in Cuba.

"What is this, a provocation?" Khrushchev wrote to US President John F. Kennedy. "One of your planes violates our frontier during this anxious time when everything has been put into combat readiness."

If the U-2's path had strayed further west, or the Soviet fighters had been fast enough to intercept it, this incident could have played out quite differently. Both the United States and the USSR had thousands of nuclear missiles ready to fire. Instead of a nearly-forgotten anecdote, the U-2 incident could have been a trigger for war, like the assassination of Franz Ferdinand.

Competition among the world's most powerful countries shapes our world today. And whether it's through future incidents like the lost U-2, or something else entirely, it's plausible that it could escalate and lead to a major, devastating war.

Is there anything you can do to help avoid such a terrible outcome? It is, of course, difficult to imagine how any one individual can hope to influence such world-historical events. Even the most powerful world leaders often fail to predict the global consequences of their decisions.

But I think the likelihood and severity of great power war makes this among the most pressing problems of our time. The prospect of international competition in advanced AI may increase the peril. And some opportunities to lower the risk could be highly promising to work on.

In short, there are actions we can take to navigate a particularly perilous time – and perhaps even reduce the threat of great power war for future generations.

Summary

Economic growth and technological progress have **bolstered** the **arsenals** of the world's most powerful countries. That means the next war between them could be far worse than World War II, the deadliest conflict humanity has yet experienced.

Could such a war actually occur? We can't rule out the possibility. Technical **accidents** or diplomatic **misunderstandings** could spark a conflict that **quickly escalates**. International **tension** could cause leaders to decide they're better off **fighting** than **negotiating**. And **rapidly** advancing **AI** may change the way wars are fought and lead to dangerous **geopolitical upheaval**.

It seems hard to make progress on this problem. It's also less neglected than some of the problems that we think are most pressing. There are certain issues, like making nuclear weapons or military artificial intelligence systems safer, which seem promising — although it may be more impactful to work directly on reducing risks from AI, bioweapons or nuclear weapons directly. You might also be able to reduce the chances of misunderstandings and miscalculations by developing expertise in one of the most important bilateral relationships (such as that between the United States and China).Finally, by making conflict less likely, reducing competitive pressure, and improving international cooperation, you might be helping to reduce other risks, like the chance of future pandemics.

Our overall view

Recommended

Working on this issue seems to be among the best ways of improving the long-term future we know of, but all else equal, we think it's less pressing than our highest priority areas (primarily because it seems less neglected and harder to solve).

Scale

There's a significant chance that a new great power war occurs this century.

Although the world's most powerful countries haven't fought directly since World War II, war has been a constant throughout human history. There have been numerous close calls, and several issues could cause diplomatic disputes in the years to come.

These considerations, along with forecasts and statistical models, lead me to think there's about a one-in-three chance that a new great power war breaks out in roughly the next 30 years.

Few wars cause more than a million casualties and the next great power war would probably be smaller than that. However, there's some chance it could escalate massively. Today the great powers have much larger economies, more powerful weapons, and bigger military budgets than they did in the past. An all-out war could kill far more people than even World War II, the worst war we've yet experienced.

Could it become an existentially threatening war — one that could cause human extinction or significantly damage the prospects of the long-term future? It's very difficult to say. But my best current guess is that the chance of an existential catastrophe due to war in the next century is somewhere between 0.05% and 2%.

Neglectedness

War is a lot less neglected than some of our other top problems. There are thousands of people in governments, think tanks, and universities already working on this problem. But some solutions or approaches remain neglected. One particularly promising approach is to develop expertise at the intersection of international conflict and another of our top problems. Experts who understand both geopolitical dynamics and risks from advanced artificial intelligence, for example, are sorely needed.

Solvability

Reducing the risk of great power war seems very difficult. But there are specific technical problems that can be solved to make weapons systems safer or less likely to trigger catastrophic outcomes. And in the best case, working on this problem can have a leverage effect, making the development of several dangerous technologies safer by improving international cooperation and making them less likely to be deployed in war.

At the end of this profile, I suggest five issues which I'd be particularly excited to see people work on. These are:

- Developing expertise in the riskiest bilateral relationships
- Learning how to manage international crises quickly and effectively and ensuring the systems to do so are properly maintained
- Doing research to improve particularly important foreign policies, like strategies for sanctions and deterrence¶ Improving how nuclear weapons and other weapons of mass destruction are governed at the international level
- Improving how such weapons are controlled at the national level

Why might preventing great power war be an especially pressing problem?

A modern great power war — an all-out conflict between the world's most powerful countries — could be the worst thing to ever happen to humanity.¶ Historically, such wars have been exceptionally destructive. Sixty-six million people died in World War II, likely the deadliest catastrophe humanity has experienced so far

Since World War II, the global population and world economy have continued to grow, nuclear weapons have proliferated, and military technology has continued to advance. This means the next world war could be even worse, just as World War II was much deadlier than World War I.

It's not guaranteed that such a war will break out. And if it does, it may not escalate to such a terrible extent. But the chance can't be ignored. In fact, there are reasons to think that the odds of World War III breaking out this century are worryingly high.

A modern great power war would be devastating for people alive today. But its effects could also persist long into the future. That's because there is a substantial chance that this century proves to be particularly important. Technologies with the potential to cause a global catastrophe or radically reshape society are likely to be invented. How we choose to develop and deploy them could impact huge numbers of our descendants. And these choices would be affected by the outcomes of a major war.

To be more specific, there are three main ways great power conflict could affect the long-term future:

- High international tension could increase other risks. Great power tensions could make the world more dangerous even if they don't lead to war. During the Cold War, for example, the United States and the USSR never came into direct conflict but invested in bioweapons research and built up nuclear arsenals. This dynamic could return, with tension between great powers fueling races to develop and build new weapons, raising the risk of a disaster even before shots are fired.
- War could cause an existential catastrophe. If war does break out, it could escalate dramatically, with modern weapons (nuclear weapons, bioweapons, autonomous weapons, or other future technologies) deployed at unprecedented scale. The resulting destruction could irreparably damage humanity's prospects.
- War could reshape international institutions and power balances. While such a catastrophic war is possible, it seems extremely unlikely. But even a less deadly war, such as another conflict on the scale of World War II, could have very long-lasting effects. For example, it could reshape international institutions and the global balance of power. In a pivotal century, different institutional arrangements and geopolitical balances could cause humanity to follow different long-term trajectories.

The rest of this profile explores exactly how pressing a problem great power conflict is. In summary:

- Great power relations have become more tense. (More.)
- Partly as a result, a war is more likely than you might think. It's reasonable to put the probability of such a conflict in the coming decades somewhere between 10% and 50%. (More.)
- If war breaks out, it would probably be hard to control escalation. The chance that it would become large enough to be an existential risk cannot be dismissed. (More.)
- Competition over AI may increase the chance of conflict, and advanced AI systems may make any resulting war more likely to be devastating. (More.)
- This makes great power conflict one of the biggest threats our species currently faces. (More.)
- It seems hard to make progress on solving such a difficult problem (more) — but there are many things you can try if you want to help (more).

International tension has risen and makes other problems worse

Imagine we had a thermometer-like device which, instead of measuring temperature, measured the level of international tension.2 This 'tension metre' would max out during periods of all-out global war, like World War II. And it would be relatively low when the great powers3 were peaceful and cooperative. For much of the post-Napoleonic 1800s, for example, the powerful European nations instituted the Concert of Europe and mostly upheld a continental peace. The years following the fall of the USSR also seem like a time of relative calm, when the tension metre would have been quite low.4

How much more worried would you be about the coming decades if you knew the tension metre would be very high than if you knew it would be low? Probably quite a lot. In the worst case, of course, the great powers could come into direct conflict. But even if it doesn't lead to war, a high level of tension between great powers could accelerate the development of new strategic technologies, make it harder to solve global problems like climate change, and undermine international institutions.

During the Cold War, for instance, the United States and USSR avoided coming into direct conflict. But the tension metre would still have been pretty high. This led to some dangerous events:

- A nuclear arms race. The number of nuclear warheads in the world grew from just 300 in 1950 to over 64,000 in 1986.
- The development of new bioweapons. Despite signing the Biological Weapons Convention in 1972, the search for military advantages motivated Soviet decision makers to continue investing in bioweapon development for decades. Although never used in combat, biological agents were accidentally released from research facilities, resulting in dozens of deaths and threatening to cause a pandemic.5
- Nuclear close calls. Military accidents and false alarms happened regularly, and top decision makers were more likely to interpret these events hostilely when tensions were high. On several occasions it seems the decision about whether or not to start a nuclear war came down to individuals acting under stress and with limited time.

This makes international tension an existential risk factor. It's connected to a number of other problems, which means reducing the level of international tension would lower the total amount of existential risk we face.

The level of tension today

Recently, international tension seems to have once again been rising. To highlight some of the most salient examples:

- China-United States relations have deteriorated, leading to increased competition in trade, geopolitical influence, and AI technology.
- Russia's invasion of Ukraine has resulted in an estimated one million people killed or wounded, raised the risk of nuclear war, and sent United States-Russia relations to their lowest point since the Cold War.
- Chinese and Indian soldiers fought deadly skirmishes along their countries' disputed border in 2020–21.These dynamics raise an important question: how much more dangerous is the world given this higher tension than it would be in a world of low tension?

I think the answer is quite a bit more dangerous — for several reasons. First, international tension seems likely to make technological progress more dangerous. There's a good chance that, in the coming decades, humanity will make some major technological breakthroughs. We've discussed, for example, why one might worry about the effects of advanced artificial intelligence systems or biotechnology. The level of tension could strongly affect how these technologies are developed and governed. Tense relations could, for example, cause countries to neglect safety concerns in order to develop technology faster.6

Second, great power relations will strongly influence how nations do, or do not, cooperate to solve other global collective action problems. For example, in 2022, China withdrew from bilateral negotiations with the United States over climate action in protest of what it perceived as American diplomatic aggression in Taiwan. That same year, efforts to strengthen the Biological Weapons Convention were reportedly hampered by the Russian delegation after their country's invasion of Ukraine raised tensions with the United States and other western countries.

And third, if relations deteriorate severely, the great powers could fight a war.

How likely is a war?

Wars are destructive and risky for all countries involved. Modern weapons, especially nuclear warheads, make starting a great power war today seem like a suicidal undertaking.

But factors like the prevalence of war throughout history, the chance that leaders make mistakes, conflicting ideologies, and commitment problems, make me think that conflict could break out anyway.

On balance, I think such an event is somewhat unlikely but hardly unthinkable. To quantify this: I put the chance we experience some kind of war between great powers before 2050 at about one-in-three.7

War has occurred regularly in the past

One reason to think a war is quite likely is that such conflicts have been so common in the past. Over the past 500 years, about two great power wars have occurred per century.8

Naively, this would mean that every year there's a 2% chance such a war occurs, implying the chance of experiencing at least one great power war over the next 80 years — roughly until the end of the century — is about 80%.9

This is a very simple model. In reality, the risk is not constant over time and independent across years. But it shows that if past trends simply continue, the outcome is likely to be very bad.

Has great power war become less likely?

One of the most important criticisms of this model is that it assumes the risk is constant over time. Some researchers have argued instead that, especially since the end of World War II, major conflicts have become much less likely due to:

- Nuclear deterrence: Nuclear weapons are so powerful and destructive that it's just too costly for nuclear-armed countries to start wars against each other.10
- Democratisation: Democracies have almost never gone to war against each other, perhaps because democracies are more interconnected and their leaders are under more public pressure to peacefully resolve disputes with each other.11 The proportion of countries that are democratic has increased from under 10% in 1945 to about 50% today.
- Strong economic growth and global trade: Global economic growth accelerated following World War II and the value of global exports grew by a factor of almost 30 between 1950 and 2014. Since war disrupts economies and international trade, strong growth raises the costs of fighting.12
- The spread of international institutions: Multilateral bodies like the United Nations General Assembly and Security Council promote diplomatic dialogue and facilitate coordination to punish transgressors.13

It is true that we are living through an unusually long period of great power peace. It's been about 80 years since World War II. We just saw that a simple model using the historical frequency of great power wars suggests there was only a 20% chance of going that long without at least one more war breaking out. This is some evidence in favour of the idea that wars have become significantly less common.

At the same time, we shouldn't feel too optimistic.

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The numerous close calls during the Cold War suggest we were somewhat lucky to avoid a major war in that time. And a 20% chance of observing 80 years of peace is not that low.14 Structural changes might have dramatically reduced the likelihood of war. Or perhaps we've just been lucky. It could even be that technological advances have made war less likely to break out, but more deadly when it occurs, leaving the overall effect on the level of risk ambiguous. It just hasn't been long enough to support a decisive view.15

So while the recent historical trend is somewhat encouraging, we don't have nearly enough data to be confident that great power war is a thing of the past. To better predict the likelihood of future conflict, we should also consider distinctive features of our modern world.16

One might think that a modern great power war would simply be so destructive that no state leader would ever choose to start one. And some researchers do think that the destruction such a war would wreak globally makes it less likely to occur. But it would be hard to find anyone who claims this dynamic has driven the risk to zero.

First, a war could be started by accident.

Second, sometimes even prudent leaders may struggle to avoid a slide towards war.

We could blunder into war

An accidental war can occur if one side mistakes some event as an aggressive action by an adversary.

This happened several times during the Cold War. The earlier example of the wayward American reconnaissance plane shows how routine military exercises carry some escalation risk. Similarly, throughout history, nervous pilots and captains have caused serious incidents by attacking civilian planes and ships.17 Nuclear weapons allow for massive retaliatory strikes to be launched quickly — potentially too quickly to allow for such situations to be explained and de-escalated.

It is perhaps more likely, though, that an accidental war could be triggered by a technological malfunction. Faulty computers and satellites have previously triggered nuclear close calls. As monitoring systems have become more reliable, the rate at which such accidents have occurred has been going down. But it would be overconfident to think that technological malfunctions have become impossible.

Future technological changes will likely raise new challenges for nuclear weapon control. There may be pressure to integrate artificial intelligence systems into nuclear command and control to allow for faster data processing and decision making. And AI systems are known to behave unexpectedly when deployed in new environments.18

New technologies will also create new accident risks of their own, even if they're not connected to nuclear weapon systems. Although these risks are hard to predict, they seem significant. I'll say more about how such technologies — including AI, nuclear, biological, and autonomous weapons — are likely to increase war risks later.

Leaders could choose war

All that said, most wars have not started by accident. If another great power war does break out in the coming decades, it is more likely to be an intentional decision made by a national leader.

Explaining why someone might make such a costly, destructive, unpredictable, and risky decision has been called "the central puzzle about war." It has motivated researchers to search for "rationalist" explanations for war. In his 2022 book Why We Fight, for example, economist Chris Blattman proposes five basic explanations: unchecked interests, intangible incentives, uncertainty, commitment problems, and misperceptions.19

This section discusses how great power tensions may escalate to war in the next few decades. It focuses on three potential conflicts in particular: war between the US and China, between the US and Russia, and between China and India. These are discussed because each of these countries are among the world's largest economies and military spenders, and seem particularly likely to fight. At the end, I briefly touch on other potential large conflicts.

United States-China

The most worrying possibility is war between the United States and China. They are easily the world's largest economies. They spend by far the most on their militaries. Their diplomatic relations are tense and have recently worsened. And their relationship has several of the characteristics that Blattman identifies as causes of war.

At the core of the United States-China relationship is a commitment problem.

China's economy is growing faster than the United States'. By some metrics, it is already larger.20 If its differential growth continues, the gap will continue to widen between it and the United States. While economic power is not the sole determinant of military power, it is a key factor.21

The United States and China may be able to strike a fair deal today. But as China continues to grow faster, that deal may come to seem unbalanced. Historically, such commitment problems seem to have made these kinds of transition periods particularly dangerous.22

In practice, the United States and China may find it hard to agree on rules to guide their interactions, such as how to run international institutions or govern areas of the world where their interests overlap.

The most obvious issue which could tip the United States-China relationship from tension into war is a conflict over Taiwan. Taiwan's location and technology industries are valuable for both great powers.

This issue is further complicated by intangible incentives.

For the United States, it is also a conflict over democratic ideals and the United States' reputation for defending its allies.

For China, it is also a conflict about territorial integrity and addressing what are seen as past injustices.

Still, forecasts suggest that while a conflict is certainly possible, it is far from inevitable. As of 8 June 2023, one aggregated forecast23 gives a 17% chance of a United States-China war breaking out before 2035.24

A related aggregated forecast of the chance that at least 100 deaths occur in conflict between China and Taiwan by 2050 gives it, as of 8 June 2023, a much higher 68% chance of occurring.25

United States-Russia

Russia is the United States' other major geopolitical rival.

Unlike China, Russia is not a rival in economic terms: even after adjusting for purchasing power, its economy is only about one-fifth the size of the United States'.

However, Russia devotes a substantial fraction of its economy to its military. Crucially, it has the world's largest nuclear arsenal. And Russian leadership has shown a willingness to project power beyond their country's borders.

Top five countries by estimated military spending, 2021. Source: SIPRI

Russia's 2022 invasion of Ukraine demonstrated the dangers of renewed rivalry between Russia and the United States-led West. The war has already been hugely destructive: the largest war in Europe since World War II, with hundreds of thousands of casualties already and no end to the conflict in sight. And it could get much worse. Most notably, Russian officials have repeatedly refused to rule out the use of nuclear weapons.

Unchecked interests and intangible incentives are again at play here. Vladimir Putin leads a highly-centralised government. He has spoken about how his desire to rebuild Russia's reputation played in his decision to invade Ukraine.

Given their ideological differences and history of rivalry, it is reasonable to expect that the United States and Russia will continue to experience dangerous disagreements in the future. As of 8 June 2023, an aggregated forecast gives a 20% chance that the United States and Russia will fight a war involving at least 1,000 battle deaths before 2050.

China-India

India is already the world's third-largest economy. If national growth rates remain roughly constant, the size of the Indian economy will surpass that of the United States' sometime this century. India also has nuclear weapons and is already the world's third-largest military spender (albeit at a much lower level than China or the United States).

One reason to worry that China and India could fight a war is that they already dispute territory along their border. Countries that share a border, especially when it is disputed, are more likely to go to war than countries that do not. By one count, 88% of the wars that occurred between 1816 and 1980 began as wars between neighbours.26

In fact, China and India already fought a brief but violent border war in 1962. Deadly skirmishes have continued since, resulting in deaths as recently as 2020.

Forecasters agree that a China-India conflict seems relatively (though not absolutely) likely. An aggregated forecast gives a 10% chance of war before 2035.

Other dangerous conflicts

These three conflicts — United States-China, United States-Russia, and China-India — are not the only possible great power wars that could occur. Other potential conflicts could also pose existential risk, either because they drive dangerous arms races or see widespread deployment of dangerous weapons.

We should keep in mind India-Pakistan as a particularly likely conflict between nuclear-armed states and China-Russia as a potential, though unlikely, conflict between great powers with a disputed border and history of war. Plus, new great powers may emerge or current great powers may fade in the years to come.

While I think we should prioritise the three potential conflicts I've highlighted above, the future is highly uncertain. We should monitor geopolitical changes and be open to changing our priorities in the future.

A race for AI might increase the risk of conflict

One major risk factor for great power conflict is the accelerating development of AI.

As companies push to create AI systems that can augment and replace much of human labour, the great powers have not been oblivious. Political leaders in the United States, China, and elsewhere recognise that the technology could reshape social, economic, and military systems.27

Facing the prospect of such a transformative technology, the leaders of great powers may feel they have more to fear from one another. They may come to believe that another great power leading in AI poses an unacceptable threat — one that justifies drastic action to defang.

There are at least three reasons falling behind a geopolitical rival in AI could seem so concerning:

- AI could enhance economic strength — with AI advanced enough to perform most occupations currently done by humans, a country could accelerate economic growth at an unprecedented rate. Since economic strength provides a major advantage in conflict, this *could* upset the existing balance of power.
- AI could enhance military might — advanced AI may also greatly strengthen a country's ability to fight wars by, for example, allowing it to develop new weapons and supporting technologies like targeting systems, improve its intelligence-gather and logistics systems, or even significantly automate its armed forces. Even more specific capabilities, such as the ability to track or disable a rival's nuclear arsenal, would be highly destabilising.
- AI could otherwise enhance geopolitical power — AI might be used to advance a country's interests in other ways, such as by strengthening its cultural dominance or helping it gain allies through strategic sharing of the technology's benefits.

A country that fears its rival might be on the cusp of achieving such major advances may feel incentivised to strike before they materialise. Or it may feel the need to race forward in its own pursuit of the technology, despite the inherent risks. The country's leaders may feel that if they wait, they'll end up in a position completely unable to compete with an opponent and subject to their domination.

There's already some evidence of tension between the United States and China over AI technology. Most notably, the United States has enacted a series of export controls to reduce China's access to advanced AI chips.

Despite the risk, it's far from certain that these dynamics must escalate. There are genuine opportunities for international cooperation on AI, which could help reduce the risk of conflict as well as the other serious risks posed by the technology. Finding shared interests and positive-sum agreements may allow the world to mitigate the dangers while increasing prosperity for all.

Overall predictions

Below is a table listing relevant predictions from the forecasting platform Metaculus, including the number of predictions made, as of 10 March 2023. Note the different timescales and resolution criteria for each question; they may not be intuitively comparable.

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I have previously independently estimated the likelihood of seeing a World War II-like conflict this century. My calculation first adjusts historical base rates to allow for the possibility that major wars have become somewhat less likely, and uses the adjusted base rate to calculate the probability of seeing a war between now and 2100.

This method gives a 45% chance of seeing a major great power war in the next 77 years. If the probability is constant over time then the cumulative probability between now and 2050 would be 22%. This is aligned with the Metaculus predictions above.

We can also ask experts what they think. Unfortunately, there are surprisingly few expert predictions about the likelihood of major conflict. One survey was conducted by the Project for the Study of the 21st Century. The numbers were relatively aligned with the Metaculus forecasts, though slightly more pessimistic. However, it seems a mistake to put too much stock in this survey (see footnote).28

We now have at least a rough sense of a great power war's probability. But how bad could it get if it occurred?

A new great power war could be devastating

At the time, the mechanised slaughter of World War I was a shocking step-change in the potential severity of warfare. But its severity was surpassed just 20 years later by the outbreak of World War II, which killed more than twice as many people.

A modern great power war could be even worse.

How bad have wars been in the past?

The graph below shows how common wars of various sizes are, according to the Correlates of War's Interstate War dataset.28

The x-axis here represents war size in terms of the logarithm of the number of battle deaths. The y-axis represents the logarithm of the proportion of wars in the dataset that are at least that large.

Using logarithms means that each step to the right in the graph represents a war not one unit larger, but 10 times larger. And each step up represents a war that is not one unit more likely, but 10 times more likely.

What the graph shows is that wars have a heavy tail. Most wars remain relatively small. But a few escalate greatly and become much worse than average.¶ Of the 95 wars in the latest version of the database, the median battle death count is 8,000. But the heavy tail means the average is 334,000 battle deaths. And the worst war, World War II, had almost 17 million battle deaths.30

The number of battle deaths is only one way to measure the badness of wars. We could also consider the proportion of the population of the countries involved who were killed in battle. By this measure, the worst war since 1816 was not World War II. Instead, it's the Paraguayan War of 1864–70. In that war, 30 soldiers died for every 1,000 citizens of the countries involved. It's even worse if we also consider civilian deaths; while estimates are very uncertain, it's plausible that about half of the men in Paraguay, or around a quarter of the entire population, was killed.32

What if instead we compared wars by the proportion of the global population killed? World War II is again the worst conflict since 1816 on this measure, having killed about 3% of the global population. Going further back in time, though, we can find worse wars. Ghengis Khan's conquests likely killed about 9.5% of people in the world at the time.

The heavy tail means that some wars will be shockingly large.33 The scale of World War I and World War II took people by surprise, including the leaders who initiated it.

It's also hard to know exactly how big wars could get. We haven't seen many really large wars. So while we know there's a heavy tail of potential outcomes, we don't know what that tail looks like.

That said, there are a few reasons to think that wars much worse than World War II are possible:

- We're statistically unlikely to have brushed up against the end of the tail, even if the tail has an upper bound.
- Other wars have been deadlier on a per-capita basis. So unless wars involving countries with larger populations are systematically less intense, we should expect to see more intense wars involving as many people as World War II.
- Economic growth and technological progress are continually increasing humanity's war-making capacity. This means that, once a war has started, we're at greater risk of extremely bad outcomes than we were in the past.

So how bad could it get?

How bad could a modern great power war be?

Over time, two related factors have greatly increased humanity's capacity to make war. 34

First, scientific progress has led to the invention of more powerful weapons and improved military efficiency.

Second, economic growth has allowed states to build larger armies and arsenals.

Since World War II, the world economy has grown by a factor of more than 10 in real terms; the number of nuclear weapons in the world has grown from basically none to more than 9,000, and we've invented drones, missiles, satellites, and advanced planes, ships, and submarines.

Ghengis Khan's conquests killed about 10% of the world, but this took place over the course of two decades. Today that proportion may be killed in a matter of hours.

First, **nuclear weapons** could be used.

Today there are around 10,000 nuclear warheads globally.35 At the peak of nuclear competition between the United States and the USSR, though, there were 64,000. If arms control agreements break down and competition resurges among two or even three great powers, nuclear arsenals could expand. In fact, China's arsenal is very likely to grow — though by how much remains uncertain.

Many of the nuclear weapons in the arsenals of the great powers today are at least **10 times more powerful** than the atomic bombs used in World War II.36 Should these weapons be used, the consequences would be catastrophic.

By any measure, such a war would be by far the most destructive, dangerous event in human history, with the potential to cause billions of deaths.

The probability that it would, on its own, lead to humanity's extinction or unrecoverable collapse, is contested. But there seems to be some possibility — whether through a **famine caused by nuclear winter**, or by **reducing humanity's resilience** enough that something else, like a catastrophic pandemic, would be far more likely to reach **extinction**-levels (read more in our problem profile on nuclear war).

Nuclear weapons are **complemented** and **amplified** by a variety of other modern military technologies, including improved missiles, planes, submarines, and satellites. They are also not the only military technology with the potential to cause a global catastrophe — **bioweapons**, too, have the potential to cause massive harm through **accidents** or **unexpected** effects.

What's more, humanity's war-making capacity seems poised to further increase in the coming years due to technological advances and economic growth. Technological progress could make it cheaper and easier for more states to develop weapons of mass destruction.

In some cases, political and economic barriers will remain significant. Nuclear weapons are very expensive to develop and there exists a strong international taboo against their proliferation.

In other cases, though, the hurdles to developing extremely powerful weapons may prove lower.

Improvements in biotechnology will probably make it cheaper to develop bioweapons. Such weapons may provide the deterrent effect of nuclear weapons at a much lower price. They also seem harder to monitor from abroad, making it more difficult to limit their proliferation. And they could spark a global biological catastrophe, like a major — possibly **existentially** catastrophic — **pandemic**.

Artificial intelligence systems are also likely to become cheaper as well as more powerful. It is not hard to imagine important military implications of this technology. For example, AI systems could control large groups of lethal autonomous weapons (though the timeline on which such applications will be developed is unclear). They may increase the pace at which war is waged, enabling rapid escalation outside human control. And AI systems could speed up the development of other dangerous new technologies.

Finally, we may have to deal with the **invention** of other weapons which we **can't** currently **predict**. The feasibility and danger of nuclear weapons was unclear to many military strategists and scientists until they were first tested. We could similarly experience the invention of destabilising new weapons in our lifetime.

What these technologies have in common is the potential to quickly kill huge numbers of people:

> A nuclear war could kill tens of millions within hours, and many more in the following days and months.
>
> A runaway bioweapon could prove **very difficult** to stop. Future autonomous systems could act with **lightning speed**, even taking humans **out of the decision-making loop** entirely.
>
> Faster wars leave **less time** for humans to **intervene**, negotiate, and find a resolution that limits the damage.

How likely is war to damage the long-run future?

When a war begins, leaders often promise a quick, limited conflict. But escalation proves hard to predict ahead of time (perhaps because people are scope-insensitive, or because escalation depends on idiosyncratic decisions).

This raises the possibility of enormous wars that threaten all of humanity.

AUKUS---1NC

Disadvantage TWO is AUKUS.

The AFF decks AUKUS's second pillar, creating a window of opportunity.

Van Der Schyff 25 [Jason Van Der Schyff, Fellow with ASPI's Cyber, Technology & Security program, "Cloud infrastructure is now critical for defence", 11/06/2025, Australian Strategic Policy Institute, https://www.aspistrategist.org.au/cloud-infrastructure-is-now-critical-for-defence/, Accessed 08/04/2026] arya

Modern **defence** capability no longer begins with submarines or fighters. It **begins** with **infra**structure. **Data centres**, subsea fibre, sovereign cloud enclaves and **globally** distributed computing **capacity** are now **essential** to how militaries **plan**, **move**, **fight** and **recover**. **Hyperscale** cloud infrastructure has become as **critical** to operations as **fuel** or **communications**. It is the digital **backbone** of modern **deterrence**.

Cloud infrastructure is already **core** to **logistics**, **command** and **control**, intelligence **collection** and **targeting**. It enables rapid processing of sensor data, distributed planning and real-time coordination across allied systems. Cloud environments allow forces to access tools, models and data they did not carry into theatre and to update or re-task them while in contact. This is the **foundation** on which AUKUS **Pillar Two** capabilities **must** be built.

That utility brings with it new dependencies. Just a few global providers now underpin much of the cloud footprint used by allied governments and militaries. **Sovereignty** is no longer just a question of where data is stored. It is also about who controls the management **layers**, who can observe **telemetry** and **access logs**, and who has the **authority** to **operate** or **disconnect** systems under **pressure**. In a crisis, those **levers** could prove as **decisive** as any **physical platform**.

The experiences of Taiwan and Ukraine offer a glimpse of what this means in practice. Taiwan, under constant grey-zone pressure, is building resilience through distributed cloud architectures that can survive physical disruption and cyberattack. Sensitive workloads are backed up offshore, while domestic infrastructure is hardened for continuity. Ukraine, under sustained missile and cyberattack, shifted its core government systems into trusted hyperscale environments abroad. These were acts of national survival.

Australia has begun to recognise this shift. Defence's cloud strategy calls for a mix of sovereign and hyperscale solutions, seeking to combine the benefits of global scale with the assurance of local control. The challenge how is to give effect to strategy. This requires disciplined implementation. That means embedding sovereignty controls into architecture, ensuring transparency across the supply chain, applying enforceable constraints on data movement and privileged access, and designing systems to function in degraded or disconnected conditions. The most important time to test these assumptions is before failure occurs.

Too often, cloud infrastructure is still viewed as a back-office convenience rather than a front-line dependency. That distinction no longer holds. AUKUS Pillar One will work only if allied forces can share data across secure, resilient and trusted environments. Pillar Two technologies such as **AI**, **autonomy** and advanced cyber **depend** on scalable **compute** and cloud-native **architectures** to **operate**. The ability to **train**, **test** and **deploy** those tools at **speed** requires **infrastructure** that is already in **place** and able to scale when needed. In both cases, cloud infrastructure is the **enable**r. Without it, **AUKUS** will **struggle** to deliver **capability** at the **pace** the strategic environment demands.

This shift also alters the regional picture. Partners across the Indo-Pacific including Japan, South Korea, India, the Philippines and Taiwan are each making decisions about cloud infrastructure. Those decisions will determine not just their resilience but also their interoperability with Australia and other trusted partners. Supporting secure, transparent and resilient adoption of cloud technologies across the region is now a strategic investment in collective deterrence. Countries that cannot operate together in peacetime will not be able to respond together in crisis.

But capabilities rest on infrastructure. Cloud ecosystems depend on power, land and people. They require forward planning, workforce development and policy alignment. Japan's recent challenges with power constraints and zoning delays illustrate what happens when digital ambition collides with physical bottlenecks. Australia is better placed, with reliable energy, geographic advantage and strong alliances. Taking advantage of this positioning will require designating cloud infrastructure as critical and as a basis of strategic resilience—designed, governed and deployed with the same intent as any other national asset.

The US is key---scaling, investment, and experience.

DiFiore 25 [Nicholas DiFiore, Strategic Interoperability Planner at The Joint Staff, AUKUS and Allied AI: Building Trilateral Defense Capabilities Across the AI Stack, 09/15/2025, FAO Journal of International Affairs, faoajournal.substack.com/p/aukus-and-allied-ai-building-trilateral, Accessed 08/04/2026] arya

The AUKUS **nations** bring **diverse strengths** and **challenges** to the co-development of defense AI. The U.S. **leads** in **investment** and operational **experience**. It established the **Joint Artificial Intelligence Center** in **2018** to **accelerate** AI adoption. In 2022, it **elevated** AI efforts under a **Chief Digital and AI Office** – signaling AI's priority status for the Pentagon. **Increased** budget **allocations** for **defense AI**, reaching **$1.8 billion** in the FY2024 budget, reinforce this priority.[ii] Through programs like the Joint Warfighting Cloud Capability (JWCC), the U.S. **leverages** partnerships with major **tech firms** like Amazon, Google, Microsoft, and NVIDIA to **bolster** access to **cutting-edge** compute and **cloud infra**structure. Yet integration with allies and partners remains a shortfall – recent Defense Innovation Board findings called for deeper partner integration to support networked defense capabilities.[iii]

The UK has **strong** academic and commercial **foundations** in AI, including firms like DeepMind. Still, its defense AI **efforts** are on a **small**er scale. Recognizing AI's importance, its 2022 Defence Artificial Intelligence Strategy set a goal to be the "world's most effective, efficient, trusted and influential" defense organization in AI for its size. [iv] To implement this vision, it established the Defence AI Centre as a hub to champion and enable AI across the armed forces[v] and invested nearly £1 billion in AI research.[vi] However, the UK defense AI sector remains underdeveloped, facing infrastructure and workforce constraints.[vii] The UK views AUKUS as a force multiplier, offering access to larger data sets and joint experimentation opportunities.[viii]

Australia is in the **early stages** of defense AI **adoption**, guided by national data and autonomy strategies.[ix]Recent investments through the Defense Innovation Hub and Next Generation Technologies Fund have supported AI-enabled decision support and automated imagery analysis pilot projects. The Defence Science and Technology Group's AI for Decision Making Initiative sponsors research targeting operational needs.[x] Despite this momentum, Australia's **small** defense-industrial base, **limited** workforce, and **reliance** on **foreign** technology present **challenges**. Canberra views AUKUS as a critical mechanism for accessing advanced capabilities, mitigating its small industrial base, and contributing meaningfully to Indo-Pacific deterrence.[xi]

Project splitting circumvents.

Van Veen 26 [Christiaan van Veen, Adjunct Professor of Law at NYU, 1-26-2026, Netherlands: Transparency needed on Microsoft hyperscale data centre, https://www.article19.org/resources/netherlands-transparency-needed-on-microsoft-hyperscale-data-centre/]

ARTICLE 19 Europe has joined five civil society organisations calling for transparency about a **major** data centre development in Amsterdam after NRC investigation confirmed Microsoft as the sole tenant of what appears to be a hyperscale facility, despite a ban on such data centres.

The development, which consists of three 85-meter towers consuming as much electricity as all households in the nearby city of Haarlem, was **approved** despite the Netherlands' 2022 moratorium on **hyperscale data centres**. The facility received permits by **splitting** a single project into **three** separate applications.

Earlier this month, ARTICLE 19 Europe signed a joint letter with Leitmotiv, Advocates for the Future, Bits of Freedom, the critical infrastructure lab, and De Goede Zaak demanding clarity from Amsterdam municipality and North Holland province on the project's ownership structure and governance requirements.

Success reduces **Chinese confidence** in **hypersonics**, but **failure** greenlights **war**.

Treloar 25 [Natalie Treloar, fellow at the Indo-Pacific Studies Center, "Extending the Golden Dome: AUKUS Pillar 2", 2025, Global Security Review, https://globalsecurityreview.com/extending-the-golden-dome-aukus-pillar-2/, Accessed 8/4/2026] arya

AUKUS pillar **two** is **designed** to **reduce** the **significant** lead **China** has in dual-use **emerging tech**nologies. An "extended" Golden Dome approach that produces strong and resilient allies may provide greater strategic deterrence than an America alone approach.

As stronger allies contribute more to collective defense through burden sharing, this can reduce the financial and military burden on the United States. Capable allies can deter aggression and manage local conflicts, promoting regional stability without constant American intervention.

If **allies** are **seen** as **weak** and **easily overrun**, it may **undermine** the **credibility** of **alliances** and security guarantees, thereby **emboldening adversaries**. Strong allies often bring **advanced** technologies and **capabilities** that **enhance** joint operations through interoperability and innovation.

Defeating fractional orbital bombardment system (FOBS), intercontinental ballistic missiles (ICBM), submarine-launched ballistic missiles (SLBM), hypersonic glide weapons, and land-attack cruise missile threats is the focus of Golden Dome, which "proposes a multilayered defense network capable of intercepting threats during the boost, midcourse, and terminal stages of missile flight." Some analysts argue that it is easiest to target these threats in the boost/ascent phase.

Targeting the boost phase can either occur from allied territory, targeting North Korean missiles from South Korea, or the homeland. The boost phase can also be targeted from space. However, targeting from space is not without its own unique set of challenges. Either way, it will not be solved by America without its allies.

Arguably, five-eyes countries (Australia, Canada, New Zealand, United Kingdom, and United States) are already entangled in American nuclear command, control, and communications (NC3). Furthermore, the US operates in coordination with other military forces as part of broader coalition operations. Hence, extending Golden Dome to allies is not only possible, but can further leverage the AUKUS pillar two effort. Moreover, a missile defense system consists of sensors, interceptors, and command-and-control systems that work together to detect, track, and intercept incoming missiles.

These necessary components exist in AUKUS pillar two working groups like the cyber capabilities, artificial intelligence (AI) and autonomy, quantum technologies, undersea capabilities, hypersonics and counter-hypersonics, electronic warfare capabilities, innovation and information sharing, and the deep-space advanced radar capability program (DARC).
AUKUS pillar two leadership should prioritize the development of technologies and supporting systems for an effective extended Golden Dome architecture. America is unlikely to solve the problem in isolation by building a "tightly integrated system of low Earth orbit (LEO) satellites, terrestrial radar stations, directed-energy platforms, and kinetic interceptors" that senses, decides, and neutralizes incoming missiles. The linkages and opportunities for the prioritization of advanced technology development for missile defense can be found in the following summaries.

Cyber capabilities encompass both offensive and defensive operations for missile defense. **Offensive** cyber **tools** are **used** to **deter** adversaries and **disrupt** their **operations** through tactics such as cyber-reconnaissance, communication isolation, targeted strikes, and network intrusions. Defensively, military forces can prioritize robust network protection, active threat disruption, and seamless coordination across units to safeguard critical systems. Cyber operations can enhance intelligence gathering, command and control, and information warfare to shape public perception and the broader information environment. As conflict evolves, training personnel in cyber tactics and integrating machine learning for threat detection and analysis can maintain strategic advantage.

AI and autonomy can transform missile defense and military operations by enhancing efficiency, precision, and decision-making. Autonomous weapon systems are used for reconnaissance, surveillance, and combat missions. AI-driven wargaming platforms simulate real-world combat scenarios to help strategists test tactics and improve readiness. In command and control, AI supports real-time data processing and analysis. AI optimizes logistics by improving resource allocation, supply-chain management, and transportation.

In intelligence and surveillance, AI analyzes vast datasets to detect patterns and identify threats. Additionally, AI monitors threats and predicts future events. Human-machine teaming allows AI systems to collaborate with human operators, combining strengths and minimizing errors.

Quantum technologies **transform** missile defenses through **enhanced** security, operational **efficiency**, and advanced **training**. In cybersecurity, quantum-resistant cryptography is being developed to protect against the threat quantum computers pose to traditional encryption. Quantum key distribution offers highly secure communication by transmitting encryption keys through quantum channels, making interception nearly impossible.

In military operations, quantum algorithms can optimize logistics, supply chains, and battlefield strategies by analyzing complex data in real time. **Quantum computing** could process **data** in **real time**, enabling **missile defense** systems to **rapidly** analyze incoming threats, allowing for quicker decision-making and more effective interception. Additionally, quantum computing could enable **highly accurate** simulations of complex systems like **nuclear reactions** and weapon designs.

Undersea **capabilities** encompass a **wide range** of **offensive** and **defensive** functions that **contribute** to **missile defense**. Offensively, submarines and other undersea platforms can strike surface vessels, submarines, and land-based targets. They are also instrumental in inserting special forces into hostile territory for reconnaissance or sabotage missions. Undersea vehicles play a key role in mine warfare, either by laying mines or clearing minefields. On the defensive side, these platforms are vital for anti-submarine warfare, enabling the **detection** and **neutralization** of enemy **submarines**. They also support surveillance and reconnaissance efforts, gathering intelligence on enemy naval movements. Undersea systems help protect vital infrastructure such as pipelines and communication cables and help ensure safe navigation.

Hypersonic weapons and **counter-hypersonic** systems are **vital** to missile defense **operations**. Hypersonic **glide weapons** and hypersonic **cruise missiles** are designed to **strike targets** with **exceptional speed**, maneuverability, and precision while evading traditional defenses. Counter-hypersonic capabilities include **advanced sensors** and **tracking systems** like **radar** and **satellite imaging** to **detect** and **monitor** hypersonic weapons. Hypersonic **interceptors** aim to **neutralize** threats **mid-flight**, while high-power **lasers** and microwave **weapons** can **disrupt** their **guidance systems**. Effective command-and-control systems are essential for coordinating these defenses, and soft-kill measures such as cyberattacks offer **additional means** to **interfere** with **hypersonic weapons**.

Electronic warfare is fundamental for gaining military advantages in cross-domain missile defense. Electronic attack includes jamming enemy communications, radar, and navigation systems. It also includes spoofing—sending false signals to enemy forces. Electromagnetic or directed-energy weapons disable or destroy enemy assets. Electronic protection ensures secure communication through encryption and satellite links and employs electronic countermeasures (ECM) to defend against attack. Electronic counter-countermeasures are used to overcome enemy ECM and maintain operational effectiveness. Electronic support can be focused on gathering intelligence through signals interception, using sensors for surveillance and target acquisition, and detecting threats in the electromagnetic spectrum.

Confidence goes nuclear.

Weitz 21 [Richard Weitz, PhD in Political Science from Harvard University, 7-30-2021, "China's Hypersonic Missiles: Methods and Motives", Jamestown, https://jamestown.org/chinas-hypersonic-missiles-methods-and-motives/, DOA: 12-9-2025]|eti

Implications for Warfighting and Strategic Competition¶ Hypersonic missiles launched on planes or ships can reach targets further away than equivalent ground-based systems launched from mainland bases. More importantly, they can approach a target from a wider range of locations than if launched from a land-based system, compounding their ability to evade existing BMD systems. In October 2020, an amateur video posted online appeared to suggest that the PLA was developing an air-launched HCM capable of being carried by a strategic bomber such as the H-6N (轰, Hong) missile carrier aircraft or possibly a more advanced successor (Twitter, October 17, 2020). In March of this year, the Beijing Institute of Technology published a study entitled, "Network for hypersonic UCAV swarms" which discussed how groups of future Unmanned Combat Air Vehicles (UCAVs, aka drones) could act in coordinated operations through networked sensors and communications at hypersonic speeds.[4][4]Shixun Luo, Zhongshan Zhang, Shuai Wang, Shuo Zhang, Jibo Dai, Xiangyuan Bu and Jianping An, "Network for hypersonic UCAV swarms," Science...¶ The PLA will likely employ hypersonic systems in combination with its subsonic and supersonic delivery systems. Because of their speed and unique trajectory capabilities, hypersonic missiles can, as first-strike weapons, facilitate follow-on attacks by non-hypersonic strike systems by disabling an adversary's air and missile defense systems. One PRC defense expert specifically observed that the range of the DF-17 would enable the system to reach the Terminal High Altitude Area Defense (THAAD) BMD system in South Korea and the SM-3 BMD interceptor in Japan, "which are security threats to China" (Global Times, October 1, 2019).¶ In addition to directly bolstering the PLA's warfighting capabilities, China seeks hypersonic delivery systems to weaken Washington's extended deterrence guarantees to its allies and partners in Asia. A recent report by the U.S.-China Economic and Security Review Commission notes that, "Capabilities that can credibly threaten the U.S. military also support Beijing's aim to intimidate and coerce regional states by fueling doubts about U.S. ability or willingness to intervene in a crisis."[5][5]Jacob Stokes, "China's Missile Program and U.S. Withdrawal from the Intermediate-Range Nuclear Forces (INF) Treaty," U.S.-China Economic and Security Review Commission,...¶ Conclusion¶ The PLA's hypersonic delivery Systems intensify Beijing's challenge to the U.S.' political-military primacy in the Indo-Pacific region. The effectiveness of China's hypersonic capabilities in battle is uncertain; in addition to the PLA being largely untested in combat, hypersonics are still an emerging military technology. Furthermore, we do not know whether China, like the United States, will choose to only arm its hypersonic missiles with conventional munitions, or whether Beijing will follow Moscow's lead and load some of its hypersonic delivery systems with nuclear warheads. In any case, China's novel hypersonic capabilities could, alongside the growing power of the PLA in general, make PRC decision-makers more confident about their ability to employ force successfully against the U.S. military. If Chinese leaders believe that they could employ these capabilities to preemptively destroy or disable key U.S. defense systems, then this would give Beijing greater incentive to strike first, raising the risk of escalation and war in a crisis.[6][6]John T. Watts, Christian Trotti, Mark J. Massa, "Primer on Hypersonic Weapons in the Indo-Pacific Region, Atlantic Council, August 2020, https://www.atlanticcouncil.org/wp-content/uploads/2020/08/Hypersonics-Weapons-Primer-Report.pdf.... If other Asian countries believed that China could more effectively fight U.S. forces, this could weaken the credibility of U.S. security guarantees to allies in Asia and further undermine regional stability.¶ Some of the measures that the United States and its Indo-Pacific allies are already taking to minimize damage from the PLA's traditional missile strike capability will also enhance defense against hypersonic weapons, for example passive measures that the U.S. military is pursuing to decrease the vulnerability of U.S. forward-based forces and facilities to China's ballistic and cruise missiles that include deception, dispersal, hardening, concealment, and mobility, as well as redundancy, recovery, and reconstitution.[7][7]U.S. Joint Chiefs of Staff, "Countering Air and Missile Threats," Joint Publication 3-01, April 21, 2017; validated May 2, 2018, V-15-18,... In the future, more active (and expensive) responses could encompass disrupting hypersonic data links and sensors, space-based sensors that can track missiles in the upper atmosphere, and novel technologies for interceptors. The 2019 Missile Defense Review directs that, "Moving forward, the United States, allies, and partners will pursue a comprehensive missile defense strategy that will deliver integrated and effective capabilities to counter ballistic, cruise, and hypersonic missile threats" (Office of the Secretary of Defense, January 17, 2019). But U.S. policymakers have yet to agree on the balance of funding for hypersonic weapons systems, enabling technologies, supporting research and development infrastructure, and hypersonic missile defense (CRS, updated July 9).¶

South Korea---1NC

Disadvantage THREE is SOUTH KOREA.

South Korean deterrence is reliant on American tech.

Kim 26 [Yang-Gyu Kim, Professor at National Defense University, "[New Year Special Commentary Series] ⑧ US-China Military Security Strategy Changes in 2026 and South Korea's Defense Strategy · EAI Global", 01/22/2026, EAI Global, https://global.eai.or.kr/en/reports/13482, Accessed 09/17/2026] arya

The U.S. demand for regular maintenance, repair, and overhaul (MRO) of U.S. naval vessels in South Korea implies that South Korean shipyards will become logistical bases for the U.S. Navy, and in the event of a **military conflict** between the U.S. and China within the First Island Chain, South Korea could become a **primary target** for Chinese strikes, whether it wishes to or not.

Therefore, South Korea needs to be vigilant against the 'modernization of the ROK-U.S. alliance' being solely utilized as a means to enhance U.S. deterrence capabilities in the Indo-Pacific region. In this context, the **primary objective** of strengthening the ROK military's **capabilities**, including SSNs, should be clearly defined and communicated externally as 'enhancing strategic stability on the Korean Peninsula,' rather than 'intervening in regional conflicts.' SSNs should be operated not for containment of China but to enhance South Korea's **role** and **responsibility** in deterring North Korea by constantly **tracking** and **monitoring** North Korean SLBM-carrying **submarines**. South Korea should consider ways to contribute to the development of the ROK-U.S. alliance by avoiding the 'threat of entanglement' while reducing the U.S. security burden.

Second, efforts to leverage the wartime operational control (OPCON) transfer as strategic leverage are also necessary. South Korea playing a leading role in the defense of the Korean Peninsula aligns perfectly with the defense policy direction of a potential second Trump administration, making it a welcome agenda item for the U.S. Indeed, at the 57th ROK-U.S. Security Consultative Meeting (SCM) last November, the defense ministers of both countries agreed to pursue verification of the Full Operational Capability (FOC) of the Future Combined Forces Command headquarters within the year, thereby completing Phase 2 of the Conditions-based OPCON Transition Plan and moving into the final Phase 3 Full Mission Capability (FMC) assessment (Kim Ho-jun, Lee Jeong-hyun, Kim Cheol-seon 2025/11/14). The U.S. now desires allies that can take responsibility for their own security and possess substantial capabilities for the benefit of U.S. interests. OPCON transfer is no longer a matter of principle for 'independent defense' but an unavoidable reality.

Therefore, South Korea must leverage the acceleration of OPCON transfer to **enhance** its actual capabilities and **bolster** its **deterrence** against North Korea. It is necessary to secure **maximum support** from the U.S. while building core capabilities currently lacking in the ROK military, such as Intelligence, Surveillance, and Reconnaissance (**ISR**) assets, missile defense systems, **space** and **cyber** capabilities, and the **military** application of AI. This presents an opportunity for a 'quantum leap' in South Korea's security capabilities that aligns with the U.S.'s new defense strategy while also benefiting South Korea.

It depends on hyperscalers.

Amoah 25 [Macdonald Amoah, independent researcher with interests across critical mineral supply chains, advanced manufacturing gaps, the industrial base, and geopolitical risks, "Data Centers at Risk: The Fragile Core of American Power", 11/11/2025, Foreign Policy Research Institute, https://www.fpri.org/article/2025/11/data-centers-at-risk-the-fragile-core-of-american-power/, Accessed 07/27/2026] arya

In an era of strategic competition, digital **reliability** is **deterrence**. A single **outage** in a **hyperscale facility** can **ripple** across **intelligence**, **surveillance**, and **reconnaissance** (ISR) networks, **disrupting** real-time operational efficiency. Data centers power the **digital age** and the associated server farms **underpin** everything from economic productivity to military **decision-making** and intelligence operations. Current data center infrastructure is composed of thousands of acres of servers, transformers, and cooling towers scattered across North America. However, these private-sector behemoths remain perilously exposed. In the digital era, failure to treat data and data centers as national defense assets is a strategic blind spot.

The dynamics of deterrence have evolved in the 21st century. Previously, **power** projection relied on fuel and steel; currently, it **relies** on **megawatts** and **computational throughput**. The **physical foundations** of the **cloud**, including transformers, rare earth elements, and copper, are as **essential** to national security capabilities as **shipyards** and **automobile assembly lines** previously were.

The same materials that make fighter jets, missiles, and satellites are what make artificial intelligence (AI), cloud computing, cyber warfare, and command and control possible. When supply chains for these materials falter, such as through Chinese export controls on certain minerals and rare earths, or when power grid bottlenecks delay new facilities, the US military's ability to process and act on information stalls. A targeted mineral embargo against the United States could undermine the infrastructure that supports military networks for **collection** and **dissemination** of ISR information.

<<TEXT CONDENSED, NONE OMITTED>>

For policymakers, the issue is that data centers are dual use. The Pentagon's Joint Warfighting Cloud Capability (JWCC) and Joint All Domain Command and Control (JADC2) networks utilize the same commercial power and server backbones that serve industry. A disruption in copper, transformer steel, or photonic components slows both sectors simultaneously. The boundary between civilian compute and military command has effectively vanished; resilience in one now requires resilience in the other. Cloud outages have a major impact on the economy, critical infrastructure, and military operations, meaning the security and the functioning of these data centers are a major national security issue. This article traces the hidden linkage between data centers, component materials, and military readiness. It quantifies the physical footprint of compute infrastructure and examines the emerging vulnerabilities in supply chains and construction timelines before outlining concrete policy steps to harden the ability of US digital computing power. The Mathematics of Data Centers: Power, Materials, and Build Cycles Compute power may seem infinite, but its physical limits are constrained by materials and infrastructure. Building a single 100-megawatt AI-enabled data center—a power requirement on par with a mid-sized American city like Dayton, Ohio—has massive physical requirements. Each site needs about 2,700 tons of copper for cabling and busbars, 150 to 200 tons of transformer-grade steel, and more than 50,000 tons of concrete and structural steel for the shell. Even the smallest components carry strategic weight: rare-earth magnets inside hard-disk drives, gold in server connectors, and indium phosphide in optical transceivers. These materials form America's compute base (see Figure 1). Figure 1. Material Intensity in a 100 MW AI Data Center (estimates) The estimates for material tonnage are a synthesis of data from multiple sources, including the International Copper Association (2022) for copper requirements, the Department of Energy (2024) for transformer steel, and Arup (2022) for other embodied materials. Figures for rare earth elements and gold are based on component-level mineral data from the USGS (2024). Every new data center build amplifies these demands. Their construction has already become one of the world's fastest-growing industrial sectors for copper, aluminum, and specialized steels. Yet production of critical raw materials remains concentrated in a handful of unstable and/or authoritarian countries. The mathematics are unforgiving: Small shortages in transformer steel or copper can delay data center completion by years, with cascading effects on compute availability across both the civilian economy and defense networks. This situation reflects the limitations that previously characterized naval logistics and fuel supply chains in the 20th century. In this context, computation has become a new form of logistics: Instead of moving oil to the front, it moves information to the decision-maker. And like oil, its flow depends on extraction, refining, and transport chains that are slow, exposed, and politically entangled. Power requirements amplify the problem. Each megawatt of data center load also requires an investment in grid capacity through substations, high-voltage lines, and cooling systems, which depend on the same constrained material inputs. Meanwhile, China's 2024 National Computing Network Initiative plans over 300 new dual-use data centers that explicitly combine commercial and defense functions. Conversely, the expansion of computing in the United States is constrained by transformer shortages, permitting delays, and utility congestion. The inequality lies in hardware: Those who govern material throughput dictate digital speed. This disparity poses a sustained threat to information supremacy. The American economy and military are racing to deploy AI, advanced analytics, and global sensor networks, yet current bottlenecks reside in the alloys. Each layer of digital acceleration adds an industrial burden that compounds over time. Rather than pursuing an endless build-up of capacity, the United States must also innovate its way out of material dependence. Advances in chip architecture, photonics integration, and compute efficiency, like domain-specific accelerators and co-packaged optics, can reduce the physical footprint of compute. Investing in efficiency at the transistor and algorithm level can stretch limited materials far further than adding another data hall. Emerging Vulnerabilities: Grid, Transformers, and Mineral Bottlenecks The most advanced command-and-control systems in the world still depend on transformers, specialized metals, and high-purity optics that each have globalized supply chains with slow replacement cycles. What once seemed like a commercial challenge is now a question of national security. A grid failure or material shortage that halts data center construction can lead to delays in the flow of intelligence and targeting data to combatant commands. Transformer availability sits at the center of this risk. Rising demand, aging manufacturing equipment, and global concentration of electrical-steel supply have extended delivery schedules from under a year to more than four (see Figure 2). The same pattern holds for high-voltage substation components and electrical steel, where China, India, and Japan dominate supply. This creates an invisible drag on digital expansion. Even if America builds new compute clusters, the energy infrastructure that powers them cannot keep pace. Figure 2. Lead-Time Ladder for Critical Compute Infrastructure The estimates for lead times are derived from an aggregation of data from the US Department of Energy (DOE) reports in 2023 and 2024, the North American Electric Reliability Corporation (NERC) in 2024, and private-sector analyses, including those from EPRI, the Uptime Institute, and McKinsey in 2023. The figures represent median durations and are subject to variation based on capacity and permitting jurisdiction. These constraints spill directly into defense readiness. The recent US Department of Energy Grid Supply Chain Review cautions that the transformer backlog, extending for many years, is causing systemic delays in both commercial and defense programs. The 2021 Texas power crisis illustrated how grid instability can lead to widespread digital blackouts. Should such a disruption coincide with cyber or kinetic attacks, command-and-control systems dependent on commercial clouds may fail during crucial mobilization phases. In contrast, China's State Grid Corporation manages dual-purpose "computing and command" corridors that ensure uninterrupted power supply for military intelligence fusion centers. The disparity is stark: US computing relies on market logistics, but China's computing is regarded as national defense infrastructure. Equally concerning is the geopolitical geography of these critical materials. China produces over 70 percent of the world's transformer steel, refines most rare-earth elements, and dominates the wafer-level photonics used in optical transceivers. Any export restriction or embargo could throttle global build-out of hyperscale compute, giving Beijing a lever over both economic and military networks. Even friendly bottlenecks matter: The United States has only two domestic transformer plants capable of producing the large units needed for defense-critical data centers. The combination of long industrial lead times and concentrated supply creates a slow-burn crisis. Power, not processing speed, may determine who holds the advantage in digital-age warfare.

<<PARAGRAPH BREAKS RESUME>>

From Compute to Combat Power: Why the Cloud Is a Warfighting Enabler

Modern warfighting **runs** on **compute**. Every mission, from satellite **intelligence** and maritime **surveillance** to predictive **maintenance** and AI-enabled **targeting**, depends on data flowing through **vast** server farms and secure fiber links. These networks form to support military and intelligence actions across Joint All Domain Operations (JADO). When that data flow slows, the tempo of sensing, deciding, and acting slows with it. Compute latency means combat latency.

Decision cycles that previously depended on human analysis increasingly rely on distributed computing infrastructures. The more **rapidly** a nation can receive, communicate, and integrate sensor data, the more **credible** its **deterrent** stance becomes. Throughout the conflict in **Ukraine**, distributed cloud architectures were pivotal. When Russian missile strikes targeted Ukraine's command infrastructure, Kyiv **rapidly** migrated intelligence and targeting data to **cloud-based systems**. That shift allowed operations to continue with **minimal disruption**, preserving pace and decision speed even under attack. Compute resilience enabled continuity of operations.

Compute is the new logistics for 21st-century warfare. It moves information instead of fuel, but its vulnerabilities are strikingly similar: limited surge capacity, globalized dependencies, and long resupply timelines. The United States once maintained strategic stockpiles of oil and munitions to ensure freedom of action in crisis. Today it must think the same way about the transformers, rare-earth magnets, and photonic wafers that are needed for a data server farm. Each is a "just-in-time" component that underwrites digital-age war.

Faster kill chains **require** the compute power hidden in the **algorithms** and AI at **physical server farms**, which depend on uninterrupted access to **power**, **bandwidth**, and **data**.

Reliance solves prolif, but the adverse causes it.

Bowers 21 [Ian Bowers, associate professor at the Institute for Military Operations at the Royal Danish Defence College, "Conventional Counterforce Dilemmas: South Korea's Deterrence Strategy and Stability on the Korean Peninsula", 01/01/2021, MIT Press,

https://direct.mit.edu/isec/article/45/3/7/95269/Conventional-Counterforce-Dilemmas-South-Korea-s, Accessed 09/17/2026] arya

Despite its support, the United States has sought to maintain **leverage** over South Korea and its **deterrence** strategy. One reason Washington initially accepted the South Korea counterforce plan in 2012 was its awareness that South Korea would remain dependent on U.S. capabilities for years to come. Most notably, South Korea's current **lack** of an independent **ISR** capability provides the United States with **substantial leverage** within the **alliance** and partially addresses entrapment concerns.34 In addition, South Korea will continue to require U.S. support for **several** high-technology products it **acquires**, such as the Global Hawk unmanned aerial vehicle (**UAV**), the F-35, and the Aegis Combat System. Given South Korea's **reliance** on specific U.S. components, it would be **difficult** to indigenously develop **advanced** weapon systems without U.S. consent.35

South Korea's leaders are likely keenly aware of this reliance, and somewhat paradoxically prefer to prepare its independent strategy within the alliance framework. For South Korea, the alliance acts as a blanket under which Seoul can develop the necessary capabilities with reduced risk of North Korea responding with force. It further secures access to advanced technology. In addition, South Korean leaders **realize**, while refusing to fully network their missile defense systems with the U.S. forces, that **combined forces** provide a more **credible** deterrent.36 For this reason, South Korea has bolstered coordination with the United States on missile defense and deterrence, and sought to improve interoperability.37

Nevertheless, South Korea is progressively drawing closer to its goal of obtaining an independent conventional counterforce and countervalue capability. As South Korea strengthens these capabilities, it will no longer be dependent on the United States to initiate counterforce strikes. Similarly, as South Korea's defense industries mature further, and the country continues to develop indigenous capabilities, it will be less dependent on imports and support from the United States. Thus, even if the United States still maintains significant leverage over South Korea—and even if U.S. support is still strongly preferable—South Korea's reliance on the United States is declining. South Korea is thus approaching a point where it could employ its strategy without U.S. support.

short-term hedging: a minimally credible deterrent

In a scenario where the alliance has ended, U.S. forces have left the peninsula, or the United States is unwilling to commit to the use of force, South Korea's counterforce capabilities would provide a stopgap deterrent. The strategy may stay the hand of North Korean leaders—or at least raise the threshold for launching a nuclear strike. Moreover, by preparing for abandonment, South Korea may lower its likelihood in the first place. South Korea's independent capability would reduce North Korea's incentives to drive a wedge into the alliance. That is, if the United States abandoned South Korea, North Korea still would not have free rein to engage in nuclear compellence threats (or nuclear blackmail) against South Korea.38

South Korea's strategy faces significant credibility challenges. Resolve is not an issue, as South Korean leaders would have compelling incentives to employ their plan to defend the homeland.39 Rather, the key problem for South Korea is the credibility of its capabilities.40 As we outline in detail below, there are daunting challenges facing South Korea's operationalization of its deterrence strategy. Its ability to preemptively destroy the North Korean arsenal and retaliate in the event of an attack are far from assured.

Nevertheless, South Korea's counterforce capability can be at least “minimally credible,” and thus have a deterrent effect. As Harrison Wagner argued in the context of U.S. counterforce targeting during the Cold War, a counter-force capability can deter “even if it is common knowledge that a counterforce attack would be a very risky option to choose—as long as it is not common knowledge that the probability that one would resort to it is zero.”41 A similar (if not identical) logic applies to South Korea's counterforce capabilities. Even if leaders in North Korea believe that a preemptive strike against its nuclear arsenal is likely to fail, they need to consider the potential for a successful South Korean attack. Because the consequences would be grave, even a remote possibility of a disarming strike might deter North Korean leaders.

Similarly, South Korean threats to retaliate and kill North Korean leaders are minimally credible, as the mere possibility of such strikes may be enough to instill uncertainty. South Korea's retaliatory strategy resembles nuclear strategies premised on “first-strike uncertainty,” which rests on a state's ability to make its opponent uncertain about whether it could prevent retaliation after a nuclear attack. Although less effective than a secure second-strike capability, such capabilities nevertheless may have a deterrent effect.42

In addition to a deterrent against North Korea, South Korea's advanced capabilities likely are intended to deter other regional actors, most notably China. Although Seoul could not hope to defeat China in a conventional conflict, the potential utility of its advanced conventional capabilities for high-impact precision strikes provides Seoul with a limited deterrence by punishment capability.43 This is a high-risk strategy when facing a superior opponent that can maintain escalation dominance. However, South Korea's ability to hit high-value targets may deter Beijing from escalating a low-stakes conflict or crisis for fear of incurring costs disproportionate to possible gains.

long-term hedging: nuclear latency

South Korea's advanced conventional capabilities may also provide a longer-term hedge by bolstering its nuclear latency. If the U.S. alliance **broke** down, South Korea would **arguably** be likely to seek to develop **nuclear weapons**. Although it is difficult to prove that such considerations have informed decision-makers, the capabilities that South Korea is acquiring—particularly, its **ballistic** and **cruise** missiles—will significantly **ease** the path to a credible nuclear **deterrent**.

Prolif escalates.

Hal **Brands 25**. 10/29/2025. Distinguished Professor of Global Affairs at the Johns Hopkins School of Advanced International Studies, “Should America Encourage Nuclear Proliferation in Asia?” RSIS. https://rsis.edu.sg/rsis-publication/rsis/should-america-encourage-nuclear-proliferation-in-asia/

For these reasons, managed **prolif**eration might make the best of a worsening situation. But there are many reasons to worry that it could just make a **mess**.

First, there is the **transition problem**. Countries that are **threatened** by a **Japanese**, **South Korean**, or **Taiwanese** bomb won't just **wait** for them to cross the nuclear finish line – they may **preempt** those programmes militarily. **Taiwan**, which most needs nuclear weapons, will **never** get them without inviting a **devastating** Chinese strike. An effort to bring about managed proliferation might **cause** the very war that the strategy is **meant** to avoid.

Second, even if new entrants cross the nuclear finish line, **small arsenals** are **vulnerable** and create **temptations** for **preemption** in a crisis. In a showdown over the **Senkaku**/Diaoyu Island**s** in the East China Sea, China might try to **eliminate** a **modest** Japanese nuclear arsenal. That **threat**, in turn, incentivises **hair-trigger launch** postures that **intensify** instability. The fact that the **geography** of Northeast Asia is relatively **compact** and missile **flight times** are relatively **short** could have the **same effect**.

Third, nuclear arsenals may not do much good in the **grey zone**. NATO's nuclear guarantees haven't deterred Russian hybrid warfare against Europe. Likewise, there's no guarantee a Taiwanese arsenal would prevent China from harassing outlying islands. Nuclear use has to be **credible**, and it's doubtful that vulnerable countries would use nuclear weapons against a better-armed adversary in anything less than an existential crisis.

Fourth, proliferation may **not** stay limited. In an ideal scenario, only a few responsible democracies would acquire nuclear weapons. But what if the dominoes **don't stop there**? Why wouldn't **Indonesia** or **Vietnam** eventually want nuclear weapons? Once East Asia becomes a proliferation hotspot, perhaps other regions, like the **Persian Gulf**, might **follow**.

Fifth, more nuclear states mean more **nuclear crises**, with **little guarantee** that America can **remain aloof**. In a **nuclear**-armed Asia, **every crisis** between Japan and China, China and Taiwan, South and North Korea, or even South Korea and Japan will take on – explicitly or implicitly – a **nuclear** dynamic. And a nuclear conflict would have global implications, even if the United States was not originally involved.

Independently, war is existential.

Alexander **Gillespie 17**, 08-01-2017, Professor of international law at the University of Waikato, "What are Trump's options on North Korea?," Al Jazeera, https://www.aljazeera.com/indepth/opinion/2017/08/trump-north-korea-170802105652010.html, accessed 12-16-25] rct shua + arjun

If war **breaks out** today, it will progress much **quicker** than the last one. As **North Korea** is in the close **proximity** of several major **population centres**, the fear of **overwhelming** losses will demand a military response within less than **20 minutes**.

If **weapons of mass destruction are used** in the chaos that follows, the death toll would grow exponentially. Anything above a couple of dozen **nuclear weapons** could turn a regional conflict into a **nuclear winter**, which would have extremely dire consequences for the planet. If **China**, which is linked to **North Korea** via a treaty of mutual aid and cooperation joins the conflict, the **World War III** may **break out**. Such a scenario can even lead to the **extinction** of our species.

Imbalance---1NC

Disadvantage FOUR is IMBALANCE.

US-China military AI integration is on a trajectory of sustained parity.

Danilov 26 [Vitaly A. Danilov, Associate Professor, Department of Theory and History of International Relations at RUDN University, 2026, A Comparative Analysis of the United States and China's Experience in the Military Use of Artificial Intelligence Technologies, RUDN University, https://journals.rudn.ru/international-relations/article/view/51302/27412 //Lunde]

Abstract. The progressive economic development of the People's Republic of China (PRC) with its further integration into the global economy, as well as the growing complexity of the international political environment due to China's rising power, have revealed the need to expand the capabilities of the People's Liberation Army (PLA) to protect China's **national interests** in conflicts with developed states. A key technological innovation of the current stage of scientific and technological progress is artificial intelligence (**AI**). AI has already entered the list of technologies that are fundamentally changing the nature of war and peace, and has the potential to fundamentally reshape the militaryeconomic power of states, as it is actively used in the most advanced fields, including the space industry and virtually all military domains, particularly in command and control systems. The main rivals for leadership in AI are the **United States** and **China**. Both countries are actively developing AI-based technological solutions to modernize their armed forces. This could lead to increased strategic competition between the two powers. The study analyzes the experience of modernizing the armed forces and the concepts of their employment by the United States and China in the context of the military application of AI. The aim of the study is to determine the impact of AI adoption on the changing nature of warfare and the strategic balance of power between these countries. The object of the research is the armed forces of the United States and China in the context of their technological modernization and institutional improvement. The authors analyze the conceptual documents of the two countries and examine changes in the organizational structure of their armed forces in light of national strategies for the development and implementation of AI technologies in the military sphere. The authors conclude that the use of AI can significantly accelerate military operations, change concepts of warfare and the strategy for employing military forces, increasing the role of autonomous systems. The PLA is acutely aware of the importance of adapting to and benefiting from advances in AI, fearing a "generation gap" between its capabilities and those of the U.S. armed forces, which are perceived as a powerful adversary and, therefore, a key benchmark for comparison. Currently, in terms of the number of weapons systems under development using advanced technologies, the two states **maintain parity**. The future trajectories of these technologies in the PRC and the United States will influence the future strategic balance of power, potentially leading to the emergence of new forms and methods of employing armed forces, as well as changes in the fundamental documents of military development.

Introduction China's rise to becoming the world's largest economy, a leading industrial power, and a competitor to the United States for global leadership has heightened demands on its armed forces as a tool for solving political problems through military means. Military modernization, which started in the People's Republic of China (PRC) in the second half of the 1990s, reached a significant milestone in 2020: the People's Liberation Army (PLA) completed its mechanization process and made significant progress in informatization and improved strategic capabilities.

The strategic importance of artificial intelligence (AI) in the global arena is undeniable, leading to intense competition not only between companies but also to a kind of 'arms race' among leading countries. The words spoken by Russian President Vladimir Putin in 2019 are beginning to sound prophetic: "If someone can secure a monopoly in artificial intelligence, the consequences are clear to all—they will become the ruler of the world."1 AI will become the cornerstone of global competitiveness in the coming decades, providing leaders in this field with significant economic and military advantages.

In the ongoing struggle for leadership in AI, China and the United States are enshrining their strategic priorities in a number of key documents. To determine how technological change is being translated into the military spheres of the United States and China, three questions must be addressed.

First, how is the use of AI technologies influencing each country's evolving understanding of the nature of warfare?

Second, how are states incorporating this technology into their military structures? It is important to consider that the challenges of organizational adoption and implementation of technological innovations are closely linked to effectiveness. These challenges are crucial for determining how innovation will impact international policy. Decades of research have shown that the impact of technological change on global politics—whether changes in the economy, society as a whole, diplomacy, or military power—depends far more on how governments and organizations decide to implement and utilize new capabilities than on the technologies themselves (Pinch & Bijker, 1984).

Third, in which areas of military affairs research is being conducted to incorporate AI technology into weapons or combat support systems?

The issues addressed in this study are partially reflected in the works of scholars such as L.Y. Hunter (Hunter et al., 2023), E. Kania (2022), J. Johnson (2021), K. Zhu and K. Long (Zhu & Long, 2019), and others. In the works of these researchers, the applied implementation of AI technologies in the military sphere in the United States and China is considered from the perspective of China's gradual replacement of the US position in this area. Unlike these studies, this research focuses not on documenting a shift in the balance of power between the two powers, but on a comparative analysis of the conceptual and institutional mechanisms for implementing AI in the military spheres of the United States and China. The authors examine this process by assessing how the characteristics of each country's national strategies, military doctrines, and organizational reforms influence the nature of the emerging strategic balance of power.

The theoretical framework for the study is neoclassical realism (Rose, 1998), which allows us to examine the complex political, diplomatic, and military foreign policy actions of the United States and China as steps by great powers seeking to realize their national interests in highly competitive international environment. To identify the specific approaches of the United States and China to the development of AI in the military sphere, the authors applied a comparative method of the two actors' modes of action based on three qualitative indicators: the reflection of the specific application of AI technologies in the military doctrines of the United States and China; the degree of implementation of AI technologies in the current military organization of the two countries; and the role of AI technologies in the promising developments of each country.

The theoretical framework for this study is neoclassical realism, as proposed by Rose (1998). This approach allows us to analyze the complex political, diplomatic, and military actions of the United States and China in terms of their efforts to pursue their national interests in a competitive international environment. To identify the specific strategies of the United States and China regarding the development of artificial intelligence (AI) in the military realm, the authors employed a comparative analysis of the two nations' approaches based on three key qualitative indicators:

1. The reflection of AI technologies in their respective military doctrines,

2. The degree of implementation of these technologies in their current military organizations,

3. The role of AI in their future military plans and strategies. Research into the development of AI in the military spheres of the United States and China inevitably carries the risk of bias, as open sources predominantly cover AI applications in the commercial sector, where transparency is higher and information is more readily available. At the same time, many military developments, their scale, and actual capabilities are not fully reflected in public reports and studies, making it difficult to objectively assess the impact of military AI. It is also impossible to objectively track which tasks are outsourced to commercial firms as part of military-civilian integration—an approach to military development that both countries adhere to. As a result, assessments of military AI are based either on limited data or indirect indicators, such as changes in doctrinal sources or official announcements of the adoption or development of new weapons using AI technologies. To partially overcome these limitations, this study employs a multi-layered approach to its source material: alongside official conceptual documents (military doctrines, national security strategies, and government programs) it also draws on data on arms procurement, institutional reforms, and verified tests of specific weapons systems. This combination of sources allows assessing not only declared intentions but also practical progress in the field of military AI. However, the reader should note that the study's findings are primarily conceptual and comparative-analytical in nature: they reflect documented trends and institutional development vectors, but cannot claim to provide a comprehensive assessment of the actual military potential of any given power.

Trends in the Application of AI in the Military Sphere

New technologies are shaping the balance of power primarily through military and economic means. On the one hand, technologies create new global political spaces that become arenas for geopolitical confrontation. On the other hand, technologies directly influence countries' ability to wage and win wars, and also indirectly affect the balance of power by strengthening a country's economic strength.

In particular, prominent researchers on the political implications of military technology developments, such as M.C. Horowitz, G.K. Allen, E.B. Kania, and P. Scharre, have noted that the success of AI in the military will depend on how easily its technologies can be replicated and integrated with commercial applications. The greater the overlap with civilian technologies, the faster companies will be able to develop similar solutions. Military AI systems specific to combat operations may retain their advantages longer. Furthermore, the success of countries in the "AI revolution" will depend on their ability to use AI to strengthen their economies and optimize their military capabilities, as well as their ability to manage the social changes brought about by these technologies (Horowitz et al., 2018, pp. 11–12).

It is important to distinguish potential technological innovations from military innovations. While military innovations often involve changes in technology, this is not always the case. Military innovations refer to changes in organizational behavior and warfighting practices of armed forces, aimed at enhancing their ability to effectively convert capabilities into power (Grissom, 2006). The use of aircraft carriers as mobile airfields by the United States and Japan during World War II (1939–1945) is a typical example: the British Navy, which relied primarily on battlecruisers and battleships, used aircraft carriers only as auxiliary assets. The United States and Japan, by contrast, appreciated their striking power and began organizing task forces of other warships around aircraft carriers. While AI has the potential to form the basis for several military innovations, it is not a military innovation itself, and no one is currently using AI in a way that could be considered a military innovation. Instead, in most cases, the use of AI in the military sphere represents the highest level of automation in the application of existing technology (Romashkina, Markov & Stefanovich, 2023, pp. 82–84).

Several potential implications for the strategic balance of power from the military use of AI should be highlighted.

First, the implementation of AI has the potential to significantly accelerate military operations. Autonomous AI systems increase the speed of decision-making and task execution, even if final decisions still rest with humans (Abaimov & Martellini, 2020, pp. 145–146). Military applications of AI are already being developed, such as Project Maven in the United States, where AI is used to analyze data and recognize images from unmanned aerial vehicles (UAVs), improving the accuracy and speed of assessments.

Secondly, the advent of AI could lead to the development of new warfare concepts (military doctrines). One such possibility is the use of swarms of small robotic platforms for military operations (Huizing et al., 2019). These systems require coordination with other military assets and are capable of adapting in real time in the face of losses. Such technologies are already commercially available, and their military application could change the structure and tactics of armed forces.

Finally, AI may also change the strategy for employing military forces, increasing the role of machines and reducing reliance on human factors. AI algorithms can help commanders optimize battle plans and manage complex operations in real time, which is especially important in modern combat, where speed of decision-making is crucial.

AI Applications in the Military: The US Military's Experience The United States views artificial intelligence as part of its Third Offset Strategy,2 which was unveiled in 2014 by Secretary of Defense, Chuck Hagel, with the goal of restoring the United States' military technological advantage.3 Therefore, the Third Offset Strategy focuses on automated control and autonomy systems, where AI plays a crucial role (Bartenev, 2016).

The first two offsets were developed in the 1950s and 1970s respectively, focusing on nuclear weapons and precision-guided missiles. The third offset builds on these earlier strategies by incorporating AI technology to enhance military capabilities.

Chuck Hagel's speech at the Reagan National Defense Forum in 2014 outlined the goals and objectives of the Third Offset, emphasizing the importance of AI and automation in maintaining US military dominance.

The first statement regarding AI was made in 2016, when the Obama administration released a roadmap emphasizing the importance of U.S. leadership in AI. However, this roadmap primarily focused on regulatory policy issues.4 That same year, the White House's Office of Science and Technology Policy established a subcommittee on Machine Learning and Artificial Intelligence (MLAI).5 The subcommittee's primary goal was to facilitate the coordination of national activities in the field of artificial intelligence. As a result of these developments, the Obama administration produced a number of documents focusing on artificial intelligence regulation and government research and development (R&D).

Human augmentation systems have been a priority for research, with a particular focus on optimizing human-AI interactions to enhance performance in various applications. Equally important goals included standardizing testing by creating accessible datasets and environments for training AI models and initiating the development of industry standards to enable objective evaluation of AI systems (Rickli & Mantellassi, 2023, pp. 16–17). It should be noted that research in these areas has long-term potential for the military.

During Donald Trump's first term, the field of AI gained further momentum in the United States. In 2017, the US Department of Homeland Security published an analytical report on AI,7 which focused on assessing the benefits and risks of the technology. That same year, for the first time, AI was included in the National Security Strategy,8 emphasizing its role in creating an information state, weapons development, and surveillance systems.

In 2018, the National Defense Strategy identified AI as a key technology for winning future wars and noted threats from Chinese and Russian investments in military AI applications that could undermine the US position.9 At a White House conference, Michael Kratsios outlined the Trump administration's goals for AI: prioritizing funding for AI development, removing barriers to innovation, training the workforce, achieving military superiority, and leading internationally. A Special Committee on AI was formed to coordinate efforts in this area.10

In 2019, an Executive Order was issued that established U.S. leadership in AI as a strategic priority for the country. The budget request identified artificial intelligence, autonomous systems, and hypersonic technologies as key areas for ensuring technological dominance.11

The Biden administration has continued to focus on military AI development, releasing an updated National AI Research and Development Strategic Plan in 2023. It built on a similar document from the Obama administration, but added two significant innovations: the emphasis on expanding public-private partnerships and the proposal for international cooperation in AI research.12

The priority of AI technologies has been demonstrated by the significant investment made by the **United States** in AI development since 2019. The United States has allocated USD 328.5 billion to AI development. In 2023, US investment in this area reached USD 67.9 billion, representing a **65**.94% increase compared to 2019 spending.13 This dynamic indicates a steady increase in the priority of AI in US national strategy.

Despite a significant number of official statements and documents published by US presidential administrations, there is still a lack of a clear organizational framework for AI policy. The institutional framework for the application of AI in the US military is provided by several government agencies responsible for AI research and development, with the Department of Defense as the primary responsible agency. In June 2018, the Pentagon established the Joint Artificial Intelligence Center (JAIC). According to JAIC Director Lieutenant General John N.T. Shanahan, the JAIC plays a critical role in transitioning research and development to practical capabilities.14

Another agency responsible for artificial intelligence projects within the Department of Defense is the Defense Advanced Research Projects Agency (DARPA), which plays a leading role in artificial intelligence research and development. DARPA's annual budget exceeds USD 3 billion, allowing it to address a wide range of research challenges.15 Additionally, in 2018, DARPA announced a multi-year investment of more than USD 2 billion in new and existing programs known as the "AI Next" campaign.16

Finally, on May 10, 2018, the White House established the Ad Hoc Committee on Artificial Intelligence within the National Science and Technology Council. This committee provides advisory support to the Executive Office of the President on AI research and development priorities, and is responsible for coordinating the work of government, industry, and academia on AI research and development.

The US military is actively developing and testing various applications of artificial intelligence (AI) for military purposes, demonstrating significant achievements in this field. In 2013, the US Navy successfully performed an **autonomous** landing of the X-47B drone, and in 2015, an autonomous mid-air refueling.17 In 2016, the Pentagon demonstrated a swarm of 103 unmanned aerial vehicles that operated as a single organism; similar tests were also conducted with unmanned boats. The Project Maven, which uses AI for **reconnaissance** and **surveillance**, has been successfully used in the fight18 against ISIS.19 A significant program is the Loyal Wingman, in which unmanned aircraft are protected by manned fighters.20 Also noteworthy is the Sea Hunter, an unmanned anti-submarine warfare vessel that has completed an autonomous voyage.21 Despite these successes, the US has not yet announced tests of lethal autonomous weapons systems using AI.

As part of the "AI Next" campaign, DARPA's agenda includes the development of explainable AI (XAI), as well as autonomous systems, including swarms of unmanned aerial vehicles and ground vehicles (the OFFSET project).22 The military services are also developing their own initiatives: the US Army is testing autonomous tactical vehicles (The Ground Vehicle Autonomous Pathways, GVAP), the US Air Force is developing AI pilots to control drones (the Skyborg program), and the Navy is developing cyber defense algorithms for its autonomous vessels like the Sea Hunter.

Thus, AI has become an integral part of US military strategic plans, beginning with the Third Offset Strategy initiatives aimed at restoring military technological superiority. In recent years, a number of programs and agencies, such as the JAIC and DARPA, have been established to actively pursue AI development for military purposes. Despite significant progress in autonomous systems such as drones and reconnaissance systems, a key challenge remains the lack of coordination and an overly complex institutional structure in AI development at the national level. Current efforts, including public-private partnerships and international cooperation, are aimed at strengthening US leadership in this area. However, the future depends on the government's ability to effectively implement its plans and maintain technological superiority in the face of increasing competition from other countries.

AI Applications in the Military: China's Approach

Although the military component of China's foreign policy is not used as frequently as economic and diplomatic instruments, the reality of Chinese long-term military planning is ambitious: by 2035, the modernization of national defense and the armed forces will be largely completed, with the prospect of creating a "world-class" army from the PLA by 2049.23

To achieve these goals, China's military-political leadership has recognized that warfare is evolving from today's "information" (信息化, xinxihua) warfare to future "intellectual" (智能化息化, zhinenghua

xihua) warfare. During this transformation, artificial intelligence is expected to become as important a factor in victory on the battlefield of the future as information technology does today.24

The concept of intelligentization entails the development and implementation of artificial intelligence and the integration of interconnected technologies necessary for the development of automated command, control, communications, surveillance, intelligence, and reconnaissance (C4ISR) capabilities (Costello & McReynolds, 2018, p. 7). According to the concept of intelligentization, the integration of AI into the processes and functions of the PLA is not an end in itself. According to one PLA scholar, it "refers to the overall operational description of the armed forces' systems, consisting of people, weapons, equipment, and combat methods."25

The main premise of the new stage of theoretical development for modernizing China's military capabilities was the focus on creating a unified organizational structure at the operational-strategic level, encompassing troops of various branches and services, with the participation of all services and branches of the armed forces. The concept was to enable the PLA to conduct "integrated joint operations" (协同联合作战, xietong lianhe zuozhan) (Shou, 2013, pp. 206–207), which would create conditions for unified command and control, rapid maneuver, comprehensive strike potential, as well as operational support, logistical support, and supply for various formations. In this way, it is intended to create a "system of systems" that includes the integration of humans and machines, using AI as the "leading" or dominant (智能主导, zhineng zhudao) element that would act as a connecting link in this entire system.

Chinese military and political circles believe that a new "military revolution" is taking place globally, presenting both challenges and opportunities for China. In August 2014, the Politburo discussed new trends in global military development and the need for military innovation. Chinese President Xi Jinping emphasized the importance of this "new revolution" and called on China to accelerate military innovation to close the gap and leap forward. He emphasized the importance of developing new military theories, strategies, tactics, institutions, and governance models for waging war in the information age.26 China's science- and technology-based military renewal strategy (科技兴军, keji xingjun) is a key element of the Chinese Communist Party's (CCP) goal of building a "powerful military" (强军目标, qiangjun mubiao) in the "new era."27 Xi Jinping stressed that in an increasingly competitive international military landscape, only innovators win, and called on China to focus on advanced military technologies. He emphasized the need to develop strategic technologies and gain unique advantages in key areas.28

China's military strategy has evolved in response to new assessments of the nature of conflicts. The latest revision of the strategy, focusing on "winning local wars using information technology," was reaffirmed in the 2019 Defense White Paper. The document also discusses a new stage in the military revolution, involving the development of high-precision, stealthy, unmanned, and intelligent weapons.29

In August 2016, the Joint Staff of the Central Military Commission (CMC) of the People's Republic of China called on the PLA to actively utilize the potential of artificial intelligence for operational command, planning, decisionmaking support, and integrate advanced technologies such as big data and cloud computing into the command system for joint operations.30 In 2017, at the 19th CPC National Congress, Xi Jinping emphasized the need to accelerate the development of military "intellectualization" and improve the capabilities of joint operations based on networked information systems.31

Artificial intelligence has become a **strategic focus** for China, influencing economic development and military potential. In 2017, AI was included in the Next-Generation Artificial Intelligence Development Plan, forming the basis of the national strategy in this area.32 Chinese technology companies play a vital role in implementing this program, particularly within the **military-civilian fusion** strategy, where AI is used for command decision-making, **military calculations**, and defense technology development. In the 14th Five-Year Plan for National Informatization (2021), China's leadership emphasized the development of key technologies such as artificial intelligence and quantum computing to enhance national competitiveness and ensure security.33 At the 2nd session of the National People's Congress in March 2024, the PLA's commitment to modernization through military-civilian integration was reaffirmed, with AI highlighted as a key element of the "revolution in military affairs."34

China is making significant progress in AI development, as evidenced by key statistical indicators. According to a World Economic Forum report, the country's AI industry will exceed USD 70 billion in 2024, involving more than

4,300 companies, and investment is expected to reach USD 1.4 trillion by 2030, including through the integration of AI technologies across various industrial sectors. China's intelligent computing power reached 260 EFLOPS in 2022, achieving a compound annual growth rate of **33**.9**%**. Furthermore, AI development is driving data growth at a rate of 26% annually, from 22 ZB in 2022 to 77 ZB by 2027. The country leads in the development of basic AI models (188 registered nationally) and the deployment of industrial robots (1.7 million units, accounting for 51% of global demand in 2023).35

Regarding the institutionalization of the function of developing and applying AI in the military, the PLA military reforms initiated in 2015 led to a historic restructuring aimed at strengthening military innovation. As part of these reforms, the PLA Strategic Support Forces were established, which strengthened China's capabilities in space, cyber, electronic, and psychological warfare, and were tasked with developing capabilities in new areas of military power. While the Strategic Support Forces are responsible for implementing developments in the Armed Forces based on operational prerequisites and needs, bodies such as the Office of the Central Civil-Military Cooperation Commission, the Science and Technology Commission of the CMC, and the CMC Equipment Development Department were established to manage the development and implementation of new technological solutions, including those using AI.

The AFF puts the US behind.

Ma 26 [Jason Ma, Weekend editor for the Fortune, 5-31-2026, Data centers could help determine who wins the next war, and a shortage of compute would be 'catastrophic,' retired general says, Fortune, https://archive.is/DhFAD //Lunde]

Americans are revolting against AI **data centers** and the power plants that feed them, but they are critical to **national security**, according to retired Air Force Lt. Gen. David Deptula, who is now dean of the Mitchell Institute for Aerospace Studies.

In a Washington Post op-ed on Thursday, he argued that data infrastructure will be key to connect the Pentagon's myriad weapons, such as long-range munitions, advanced combat aircraft, space systems, missile defenses, and drones.

"Data is no longer merely a tool of commerce. It is a strategic asset," Deptula wrote. "Nearly every function in the military depends on the ability to store, move, process, secure and exploit vast quantities of data at speed and scale."

Indeed, the war in Iran demonstrated AI's ability to analyze data from across different sources and rapidly provide **targeting information**. Helped by AI-powered platforms from Palantir, the U.S. and Israel struck **thousands** of targets within the opening days of the conflict.

The dispute between the Pentagon and Anthropic over the startup's terms of use also highlighted to Pentagon leadership how much it relies on AI to **process intelligence**.

"I'm like, holy s–t, what if this software went down, some guardrail picked up, some refusal happened for the next fight like this one and we left our people at risk?" Emil Michael, the Defense Department's under secretary for research and engineering, recalled in a March episode of the All-Inpodcast.

Similarly, Deptula pointed to Ukraine's use of drones that increasingly incorporate **autonomous technology** as evidence for AI's future in warfighting.

And the fact that Iran attacked Amazon data centers in the Middle East at the beginning of the war also prove that such infrastructure is an **extension** of national power, he added.

"A shortfall in data storage and computing capacity could be catastrophic," Deptula warned. "Success in future warfare will depend on whether a belligerent has the capability to sense, decide and act faster than an adversary. That requires enormous quantities of intelligence and reconnaissance, cyber, logistics, targeting and operational data. It requires computing capacity to train AI on all of it."

While he insisted data centers are a national security priority, many Americans don't see it that way. Instead, they are pushing back against plans to build them and have made them a potent political issue.

In addition, rising electricity bills, which voters are blaming on demand from AI data centers, are also fueling the nationwide revolt.

At the same time, Deputla's plea that data centers are an essential part of the Pentagon's **arsenal** could get drowned out by messaging from President Donald Trump, who has described everything from his tariffs to his White House ballroom as crucial to national security.

The U.S. is still the global leader in data centers, for now. But Deputla pointed out that China is quickly catching up and has demonstrated the ability to mobilize its **industrial might** to build up infrastructure.

"The nation with the best data infrastructure will possess a decisive advantage in the next era of warfare," he said. "The U.S. cannot afford to **lose** that position."

It's hyperscale.

USAC 26 [U.S. Army Communications, Army's provider and maintainer of command, control, communications, 03-26-2026, Army reaches conditional agreement with private industry for hyperscaled data centers, Army, https://www.army.mil/article/291360/army_reaches_conditional_agreement_with_private_industry_for_hyperscaled_data_centers //Lunde]

WASHINGTON – The U.S. Army has conditionally selected two companies to enter into exclusive negotiations to build and operate commercial **hyperscaled data centers** on two Army installations. The initiative demonstrates a model for industry relationships under the Army's Enhanced Use Lease program.

Global Investment Firm Carlyle (NASDAQ: CG) was selected for a project on about 1,384 acres at Fort Bliss, Texas, and CyrusOne, a portfolio company jointly held by funds managed by KKR and BlackRock, was selected for a project on approximately 1,201 acres at Dugway Proving Ground, Utah. The companies were chosen through a rigorous and competitive process and will be responsible for financing, building, operating, maintaining, and decommissioning the data centers on underutilized but non-excess Army land at no upfront cost to taxpayers.

"AI is a strategic asset for the Army," said Secretary of the Army Dan Driscoll. "It is a **force multiplier**, supports future transformation and requirements, keeps the Army ahead of our adversaries, and generates resiliency across the force. These data centers are a **critical** resource to support that strategic imperative."

That erodes MAD and causes global war.

Pimpale 26 [Govind Pimpale, Research fellow at the Foundation for American Innovation, where he focuses on AI policy, 6-25-2026, An AI Capabilities Gap Can Endanger Nuclear Deterrence, https://ai-frontiers.org/articles/an-ai-capabilities-gap-can-endanger-nuclear-deterrence //Lunde]

Since the late 1950s, Mutual Assured Destruction (**MAD**) has served as a limiting factor on great-power conflict. The doctrine holds that, if two opposing nations have nuclear weapons that can survive one another's **initial strike**, then the near-certainty of **devastating retal**iation will **deter** each side from launching a

large-scale nuclear attack. Despite this logic, military planners have long considered the possibility of a counterforce nuclear attack, where a superpower uses nuclear weapons to cripple the nuclear capabilities of its enemy. If such an attack were executed preemptively, as a so-called "first strike," it could both start and end a great-power conflict in a matter of hours: without retaliatory capacity, the defender would be at the mercy of the aggressor's remaining nuclear weapons and forced to surrender.

The reason this does not happen is that a truly successful counterforce strike is nearly impossible to pull off: a would-be attacker doesn't know the locations of all opposing missile launchers and submarines, nor does it have missiles with sufficient precision and speed to destroy opposing missile silos before they could launch retaliatory nuclear warheads. Thus, if provoked by a first strike, the opposing side would likely be able to launch a large-scale nuclear response, and the attacker, unable to counter all enemy missiles, could face devastating losses. The result of a preemptive counterforce strike, in other words, would be mutual assured destruction.

AI could **change** this dynamic. By the mid-2030s, AI-assisted research and development could reduce the **cost** required to develop and manufacture military hardware by an order of magnitude. Such lower costs could enable a nation with an AI lead to quickly and cheaply build military infrastructure projects, making nuclear counterforce strikes a **realistic** possibility. Nations with weaker AI capabilities would struggle to quickly build the countermeasures needed to retain a credible nuclear deterrent.

This article will primarily be a technical assessment of how AI could undermine nuclear deterrence, although I'll lightly touch on a few political aspects. For a longer and more in-depth assessment of the factors discussed here, consider taking a look at an earlier piece I wrote.

Why MAD Has Worked So Far

In any preemptive counterforce strike scenario, the attacker is at a large disadvantage compared with the defender, because they must destroy all of the defender's nuclear weapons. To execute a counterforce strike, the attacker must achieve all the following requirements simultaneously:

Suppress launch on warning. Many nuclear-armed nations have a policy of launch on warning—launching a retaliatory strike based on sensor data, before a nuclear weapon has been confirmed to land on their territory. The attacker must either use fast-arriving weapons or otherwise disable launch on warning.

Locate and destroy nuclear submarines. Nuclear submarines, while extremely stealthy, are vulnerable once detected. Each nuclear submarine can carry and launch hundreds of warheads. The attacker must accurately track their locations in real time and destroy them all within minutes.

Locate and destroy mobile launchers. China and Russia (although notably not the US, the UK, or France) each field a set of mobile missile launchers that can be dispersed during times of high alert. These mobile launchers are extremely difficult to locate for long enough to successfully strike. However, if accurate and up-to-date position data can be provided, they are easy to destroy.

Destroy all silos. Hardened silos have known positions, but they require almost a direct hit—ideally with a nuclear weapon—to ensure their destruction. They stand out among nuclear launch platforms, as they have the fastest reaction time and the hardest-to-disrupt communications.

Defend against surviving missiles. If the attacking nation has a functioning missile defense system, it may not need to destroy all of an enemy's nuclear weapons, as its missile defense can handle some leftovers. The stronger a nation's missile defense, the less thorough its first strike has to be.

Overcome the nuclear taboo. One theory for why nuclear weapons are rarely used is the nuclear taboo. The leaders of both the US and the USSR recognized the gravity of nuclear weapons usage, and had serious humanitarian reservations about starting a nuclear conflict. Even if a counterforce strike were perfect, radioactive nuclear fallout was expected to kill tens of millions in the target country.

The six requirements above have thus far preserved deterrence not because they are physically impossible to meet but because meeting them at the scale required has always been prohibitively expensive.

A useful paradigm to consider in nuclear conflict is the cost-exchange ratio. When one side fields an additional weapon, how much must the other side spend to neutralize it? The concept comes from the Cold War debate over ballistic missile defense. If an attacking nation can build another nuclear missile for $1 million, and the interceptor needed to stop it costs the defending nation $10 million, then missile defense is a losing game: each $1 spent on offense forces a $10 expenditure on defense. This particular cost-exchange ratio explains why a nation launching a counterforce strike would struggle to defend against surviving missiles.

Why AI Could Dramatically Cut Military Infrastructure Costs

AI-assisted military R&D could invert the **cost-exchange ratio**. If a nation with an AI lead can automate most of the design, systems integration, and production labor that currently makes military infrastructure expensive, that nation's costs could be significantly reduced while its enemies' costs stayed the same (since they don't have the AI advantage). If interceptors cost only $100,000 and missiles still cost $1 million, then it suddenly becomes rational to build more interceptors.

The likelihood of this prediction's coming true rests on answers to two questions: (1) whether AI can automate most of the intellectual labor, and (2) whether doing so really cuts costs by a margin significant enough to change militaries' economic calculus. Notably, this forecast does not require progress in robotics, which could further reduce costs. I'll focus on aerospace manufacturing, which comprises most of what the attacker needs to build: new missiles, surveillance constellations, and missile defense systems.

Can AI automate most intellectual labor? Aerospace projects of the type I'm describing here are highly interdisciplinary, and even simple systems demand expertise in electrical, computer, and mechanical engineering. More-advanced projects may require fundamental research in applied physics or math. While AI assistance for software development is a relatively mature use case, AI abilities in the other fields are much more nascent.

Yet there is good reason to believe that most intellectual fields will follow the same trends. AI abilities have increased at roughly the same rate across various domains, as shown in the chart below.

Source: "How Does Time Horizon Vary Across Domains?," by Thomas Kwa and Vincent Cheng, METR (2025).

Additionally, we're already beginning to see AI labs show interest in automating science and engineering fields: Anthropic is training on electrical engineering, and the company showcased 3D modeling performance in its Claude Fable 5 launch post. OpenAI has made discoveries in mathematics and physics. On the biology front, Google DeepMind's AlphaFold has solved protein folding, enabling advances in drug discovery.

Extrapolating from such innovations, one engineer could do the work of many: outsourcing most of it to agents and handling only what agents can't yet do, like running experiments or meeting stakeholders in person.

How much can automated intellectual labor lower military R&D costs? Although aerospace manufacturing seems like a labor-heavy job, it's a remarkably white-collar profession, with high exposure to AI automation. In a 2021 article, Princeton University computer scientist Edward Felten and co-authors calculated that aerospace manufacturing has an AI Industry Exposure Score of 0.519, in the 70th percentile of all industries (around the same level as real estate). If AI automates the intellectual labor of aerospace manufacturing, this would substantially lower the cost of the final product.

To make things more concrete, consider SEC filings showing the financials of two public launch services companies, Rocket Lab and Firefly Aerospace.

SEC filing data from Rocket Lab and Firefly Aerospace.

R&D (which accounts for 28%–48% of spend at the two companies) is mostly engineering compensation and analysis, with the remainder going to propellant, test facilities, and hardware. It's mostly cognitive labor, and therefore a potential target of future AI automation. SG&A expenses (21%–22% of spend) describe the cost of finance, legal, contracts, compliance, and business development. These disciplines rely almost completely on cognitive labor. Cost of revenues (31%–51%) is the most mixed category: it blends physical labor (technicians doing welding, assembly, and launch operations) with white-collar labor (manufacturing, quality, and test engineering), and purchased materials and components.

The costs of aerospace components are themselves quite compressible. For example, star trackers (devices used by satellites to measure their own orientation) can cost over $100,000. These devices' physical components—a camera sensor, a housing, and some processing hardware—may cost as little as a few thousand dollars altogether. The more significant cost comes from qualification testing and R&D (because of low manufacturing volume, such costs can't effectively be amortized). So, by reducing the cost of R&D, automating intellectual labor can drive down component costs.

I expect the cost savings to be even greater for defense aerospace products, which usually cost more than comparable commercial products. This higher cost partly reflects the increased documentation and security practices required for highly regulated uses. Such regulatory compliance practices rest almost entirely on cognitive labor that can be automated. Another driver of these product's higher costs is costlier labor: defense contractors must have security clearances, limiting the worker pool and raising wages. An AI, on the other hand, must be cleared only once; it can then be scaled indefinitely.

What Could We Build?

Recall the six requirements that have kept a counterforce strike out of reach. Each one has held not because it was physically impossible, but because meeting it at scale was prohibitively expensive. Once cheap intellectual labor collapses those costs, an attacker with an AI lead could pursue several military megaprojects at once, each one solving a different requirement needed to launch a successful counterforce strike. This section will focus on the three requirements I expect to become significantly easier to meet.

Locating and **destroying** nuclear submarines. Research has shown that, in certain conditions, moving submarines leave a surface wake detectable by Synthetic Aperture Radar (SAR), a space-based radar system. A large constellation of SAR satellites would permit near-continuous surveillance of the entire world, regardless of time of day or local weather conditions. The US is already pursuing a SAR satellite constellation, and will launch its first satellite in 2028. The US could supplement this effort with a space-based constellation of Light Detection and Ranging (LiDAR) satellites. Each satellite would measure depth via pulses of intense light, detecting even stationary submarines (although that would require clear weather). Certain frequencies of LiDAR can detect submarines within 200 meters of the surface, within typical nuclear submarine operating depths. A dense constellation of LiDAR and SAR satellites could sweep the oceans, and expose submarines at scale.

However, while both satellite methods can reveal submarine positions temporarily, the only tool to keep track of them consistently would be underwater drones (often called Unmanned Underwater Vehicles, or UUVs). **UUVs** are currently limited by the difficulty of autonomous operation, but AI R&D would likely significantly improve this. The US has been funding research in this direction.

Defending against surviving missiles. A system with tens of thousands of interceptors prelaunched in space (similar to the 1980s Brilliant Pebbles concept, originally abandoned due to cost issues) could counter any missiles that are missed by the first strike. The US is already pursuing this, too, with its Golden Dome system.

Overcoming the nuclear taboo. The missiles of the Cold War had poor precision, so warheads with an explosive yield of hundreds of kilotons were common (designed to guarantee a silo kill). But, as mentioned, such heavy warheads would result in enough fallout to guarantee millions of deaths. In 2017, Keir A. Leiber of Georgetown and Darryl G. Press of Dartmouth found that modern missiles have much higher precision than those from the Cold War, and future improvements could reduce the average targeting error to mere meters. With such high precision, very low-yield weapons could be used, with little to no fallout. A counterforce strike could be accomplished with an estimated death toll of around tens of thousands rather than millions, well within the range of wars nations are willing to start.

Other aspects of the retaliator's deterrent are vulnerable to AI R&D too. The retaliators' mobile missile launchers can be located with the exact same SAR constellation that we discussed for submarine detection, leaving them vulnerable to a first strike. Launch on warning systems are also **vulnerable**. While it's unlikely they could be hacked outright, Anthropic's Mythos demonstrated AI driven vulnerability discovery that could be used to **confuse**, delay, or reduce confidence. Additionally, the defender's early warning response time can be shortened significantly with depressed trajectory submarine-launched missiles, which could cover 2000 km in only 10 minutes. The combination of high-speed and precise missiles also works to efficiently counter fixed silos.

The Defender's Options

If a lagging nation **realizes** that a rival is on course to achieve **nuclear primacy**—the ability to execute a counterforce strike without retaliation—it still has a few options.

The first, and most straightforward option is to expand the arsenal. It can increase the number of silos, build more nuclear submarines and mobile missile launchers, and raise its level of alert. This strategy would work in the short term, since the attacker would be forced to scale up its own forces until it was certain it could neutralize all of the new forces. The problem is the cost-exchange ratio. If the attacker has an AI advantage, the retaliator could end up paying a larger price for each new silo than the attacker pays to build the missiles or interceptors that could defeat it.

The second option is to target the root cause: the AI gap between the attacker and retaliator. Potential avenues in this direction can range from relatively diplomatic to highly **escalatory**. Options include disrupting the attacker's supply chain, launching

data poisoning attacks, or sabotaging its AI training runs. However, the most extreme actions—direct **kinetic attacks** on datacenters or researchers—would be likely to **start wars**. But even these interventions will only be **effective** if **applied early**. Once the lead is large enough, sufficiently smart AIs will already be trained. Thus, an AI-lagging defender must be alert enough to act **before** the AI capability gap becomes overwhelming.

Finally, the attacker and retaliator could negotiate a treaty. A bilateral arms-control regime could in principle cap satellite constellations, ballistic missile interceptors, or AI compute used for military R&D. There is precedent here, especially in the nuclear domain. However, the main challenge will be aligning incentives. The leading nation has no incentive to join a treaty where only the lagging party stands to gain. Traditional arms-control treaties have only worked where there were symmetric costs on both sides.

AI May Disrupt Nuclear Deterrence

In conclusion, the historical robustness of nuclear deterrence has rested on a cost-exchange ratio that favors the retaliator, but AI-assisted R&D can invert the ratio. This might enable a single superpower leading in AI to achieve nuclear primacy. The AI superpower would then wield enormous leverage, as it could credibly threaten to win just about any war. Such negotiating leverage could reshape the global balance of power, even if nuclear weapons are never used.

Even before military infrastructure megaprojects are complete, they will affect policy. If major powers come to believe that AI R&D may make their nuclear deterrents less effective, they will have incentives to expand arsenals, shorten decision timelines, rely more heavily on launch on warning, contest one another's space architectures, and target the AI and semiconductor bases that underpin their adversaries. This new equilibrium would increase military spending, shorten decision times, and raise the risk of war.

Econ---1NC

Disadvantage FIVE is ECON.

The AFF causes fast recession.

Zaman 26 [Jazib Zaman, CEO at TECHi, Former Forbes Technology Council member, 5-6-2026, AI Capex Is Now 75% of U.S. GDP Growth: Why Big Tech's Spending Boom Cannot Be Stopped, https://www.techi.com/ai-capex-carries-us-economy-token-factories/] //roro + recut Lunde

Big Tech is spending so much money building **AI infrastructure** that it is, quite literally, carrying the entire U.S. economy on its back. The capital expenditure ("capex") going into AI systems — data centers, GPU clusters, networking, power — is on track to add **2.5%** to U.S. GDP growth in 2026 and more than 3% in 2027. In Q1 2026 alone, AI-related capex was responsible for an astonishing 75% of all U.S. economic growth. The combined 2026 capex commitment from the top-five hyperscalers — Amazon, Microsoft, Alphabet, Meta, and Oracle — is now tracking above $440 billion, and that figure understates the real number because it ignores private AI labs, sovereign projects, and enterprise deployments. Stopping the boom is no longer a policy debate; it is a **recession trigger**. In the geopolitical context of USA versus China, the policy answer is settled.

The 2.5% GDP Number Is Probably Too Low

The official tally — 2.5% of 2026 GDP growth from AI capex — is a **hyperscaler-only** estimate built on top-down BEA inputs and the disclosed 2026 capex guidance from the five companies that dominate the AI infrastructure spend. It does not capture the rapidly expanding cohort of private AI labs (OpenAI, Anthropic, xAI, Mistral) buying their own clusters; it does not capture sovereign AI projects (UAE, Saudi Arabia, Japan, India) standing up national capacity; and it does not capture the enterprise deployment wave —

banks, retailers, pharma, defense — pulling forward five-year IT plans into 18-month AI rollouts. Stacking those layers, real AI capex is closer to 3.5-4% of GDP this year and could exceed 5% by 2028.

The "Token Factory" Phase vs the ROI Phase

Right now the money is going into building token factories — physical AI data centers stuffed with NVIDIA Blackwell, AMD MI400, and increasingly custom silicon (Google TPU v6, AWS Trainium 3, Microsoft Maia 200). The output of a token factory is not a service yet; it is the capacity to produce inference and training tokens. This is the canal-and-railroad phase of an industrial revolution: enormous fixed-cost spend with minimal revenue contribution at the moment of deployment.

The ROI phase begins when those factories run real workloads at scale: AI agents writing production code, automating workflows, condensing legal review, generating drug-discovery candidates, replacing customer-support headcount. The economic literature is consistent — every previous general-purpose technology platform (electricity, the internet, mobile) produced returns on capital that exceeded the build-out cost by an order of magnitude, but only after the buildout was largely complete. We are not there yet.

The capex-to-revenue gap

Microsoft disclosed roughly $13 billion in annualized AI revenue in its most recent quarter, against ~$89 billion in AI capex — a 6.8× gap. That ratio is structurally unsustainable forever, but historically it is what every infrastructure cycle looks like in years one to four. The bull case requires the gap to close from the revenue side; the bear case requires it to close from the capex side.

Why the Government Will Not Stop the Boom

Some policymakers are nervous about AI capex concentration; the FTC has flagged vertical integration; the EU continues to litigate digital markets law. None of it matters in the near term. Pulling back AI spend would mechanically print a recession headline within **two quarters**. No administration is going to volunteer for that outcome, particularly when the political framing is dominated by USA-vs-China strategic competition.

Beijing has tripled state-directed AI investment since 2024 and is racing to deploy domestic GPU alternatives at scale. Washington's read is that this is a national-security competition with a productivity dividend on the other side. That framing virtually guarantees policy support — through tax incentives, accelerated permitting for power and data centers, and continued export-control posture against advanced semiconductor sales to China. The NVIDIA position in this geopolitical equation is the single most important corporate variable.

How AI Capex Trickles Down to the IA13 Cohort

Hyperscaler capex does not vanish. Every billion that Amazon, Microsoft, Google, Meta, and Oracle spend on AI infrastructure flows directly into a narrow basket of providers — the IA13. The cohort spans GPU and accelerator silicon, networking, data-center physical layer, and power generation. Order books are committed 12-18 months out for most of these names.

Growth caps great power war, but downswings cause it.

Shuguang **Jiang 25**. Professor at the Shandong University Centre for Economic Research. 3/3/2025. “War and Peace: How Economic Prospects Drive Conflictuality.” HAL Open Science. https://hal.science/hal-04967990/file/War_and_peace_WP3March25.pdf

Although peace and development are central themes of our time, various forms of conflict – between nations, ethnic groups, organizations, and individuals– remain **pervasive**. High-profile geopolitical tensions, such as the ongoing conflicts between Russia and Ukraine and in the Middle East, serve as **stark reminders** of the preciousness of peace. The **shifting global landscape** and power struggles among major nations are particularly concerning. Thucydides’s Trap, a concept popularized by political scientist Graham T. Allison (Allison, 2015, 2017), draws from the ancient Greek historian Thucydides, who noted that the rise of a new power **often led** to conflict with an established one. The idea has gained significant attention in contemporary international relations, particularly in the context of the perceived rivalry between the **United States** and **China**.

Historical accounts underscore the recurring nature of power transitions leading to conflict. For example, the rise and fall of British naval mastery, as discussed by Kennedy (2017), and broader analyses of war and change in world politics by Gilpin (1981) illustrate the Thucydides Trap.1 Allison (2015) refers to **16** historical cases over the past 500 years where a rising power challenged an

established power, finding that **12** resulted in war. The two World Wars are also prominent historical cases. These historical perspectives highlight the potential for instability and conflict during significant power shifts.

This deterministic view has been challenged by scholars like Lee (2019) and Chan (2020) who argue that conflict is **not inevitable** and that other factors, **particularly economic** conditions, can influence the trajectory. This argument is further supported by analyses of **historical power shifts**, such as Britain's response to its relative decline (Friedberg, 2021) and the dynamics of **power transitions** in Asia (Shambaugh, 2005).2 This opens the door to investigating whether economic conditions can alter the course toward **conflict or cooperation** in power dynamics.

Economic conditions are **undeniably crucial** in determining international conflicts. World War II, for example, was significantly influenced by the **Great Depression**. Economic prospects also influence **domestic politics** and conflicts. Collier and Hoeffler (2004) suggest that economic conditions largely determine the opportunity for rebellion in civil conflicts, while Blattman and Miguel (2010) and Ray and Esteban (2017) identify lower income levels and weak economic growth as strong predictors of **civil wars**. Gartzke (2007) and Mitra and Ray (2014) highlight the role of economic development in reducing war and mitigating communal violence. The role of economic factors in power transitions is more complex. Gilpin (1981) and Kennedy (2017) argue that disparities in economic growth can disrupt the **balance of power**, potentially leading to instability and conflict.

Building on this, we explore how economic prospects affect the likelihood of cooperation and conflict between rising and established powers. We hypothesize that growth prospects encourage cooperative strategies, as both parties stand to benefit from mutual gains. Economic **interdependence** driven by positive economic prospects can foster stronger **trade** relations, **investment**, and **collaboration** in technology and infrastructure, creating a **stabilizing effect** where both powers have a **vested interest** in maintaining peace. Conversely, bleak economic prospects can intensify competition over **limited resources**. An established power may perceive the rising power's growth as a **threat to its dominance**, prompting preemptive actions. Similarly, a rising power **facing economic difficulties** may adopt **aggressive strategies** to secure resources and markets, escalating tensions.